PREDICTING MENTAL HEALTH RISKS IN VIDEO GAMERS

A COMPARATIVE STUDY OF MACHINE LEARNING, DEEP LEARNING, AND FEATURE ENGINEERING TECHNIQUES

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Abstract

With rising concerns about mental health, scalable solutions for early risk identification are increasingly sought. This thesis explores the potential of machine learning models to predict mental health risk levels among video gamers, categorized as low, moderate, or high, based on scores from the Satisfaction With Life Scale (SWLS). The study uses data from a large international survey of approximately 13,000 gamers, including psychological assessments, demographics and gaming behaviors.

Unlike prior studies, this research applies multiclass classification and compares traditional models (Logistic Regression, Random Forest, Support Vector Machine), advanced ensemble models (XGBoost, LightGBM, CatBoost), and a deep learning architecture (TabNet). A key contribution is the comparison of manual categorization and sentence embeddings for processing open-ended responses, aiming to assess how different feature representations impact predictive performance.

The results show that CatBoost (manual categorization) and Logistic Regression (sentence embeddings) achieved the highest F1 Macro scores 0.4882 and 0.4824, respectively. However, all models struggled to classify moderate-risk cases. Contrary to expectations, sentence embeddings did not outperform manual categorization.

These findings highlight the potential of machine learning for mental health risk screening but underscore the need for richer behavioral data and improved modeling strategies in future research.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

The dataset used in this thesis was obtained from the Open Science Framework (OSF) repository (Sauter & Draschkow, 2017), where it is publicly available, and the owner provides consent for its use. This study did not involve collecting data from human participants or animals. All figures and images included were created by the author. As a debugging tool to

resolve implementation issues ChatGPT (OpenAI, 2025) has been used. For grammar and spelling support, the author used Thesaurus (Thesaurus.com, 2025) and Grammarly (Inc., 2025).

2 INTRODUCTION

In recent years, concerns about psychological well-being have intensified across diverse populations. Increasing levels of stress, anxiety, and dissatisfaction with life have been reported globally, driven by factors such as social isolation, economic instability, and digital overload (Fergusson et al., 2015). As mental health becomes an increasingly pressing social challenge, researchers and practitioners alike are exploring innovative tools to detect early signs of psychological distress and enable timely interventions (Whitton et al., 2021). Within this context, subjective well-being and particularly life satisfaction, plays a critical role in identifying individuals at risk of mental health problems (Keyes, 2002).

Alongside these broader concerns, video gaming has become a global phenomenon that offers a unique context for examining mental health. On the one hand, it has been associated with positive psychological outcomes, such as stress relief, enhanced emotional resilience, and improved cognitive functioning (Bowman et al., 2022). Commercial games have also been shown to foster social bonds and emotional regulation, supporting mental health management when used appropriately (Birk et al., 2019; Kowal et al., 2021).

On the other hand, excessive gaming has been linked with increased anxiety, depressive symptoms, and compulsive gaming behaviors, leading to the formal recognition of Gaming Disorder (GD) by the World Health Organization in the 11th Revision of the International Classification of Diseases (World Health Organization, 2018). Particularly in competitive online environments, high-intensity gaming may contribute to social withdrawal and psychological distress (Kuss et al., 2012; Wei et al., 2012). Recent studies have further found that gaming motivations, such as escapism and achievement-driven play, can predict excessive engagement, increasing the risk of addictive behaviors and emotional dysfunction (Kuss et al., 2012; Sauter et al., 2021).

Identifying at-risk individuals in this domain is therefore essential for designing effective intervention strategies especially as gaming communities continue to grow. Improving predictive models for mental health risk identification is critical for supporting early interventions.

While machine learning (ML) methods are increasingly used to predict mental health outcomes, much of the existing research concentrates on general populations and small datasets (Jemili, Korbaa, et al., 2024; Serin

et al., 2010). Within the gaming community, mental health research has primarily focused on correlations between gaming behaviors and psychological well-being, rather than developing predictive models (Hazel et al., 2022; Johannes et al., 2021). When predictive approaches are employed, they often rely on traditional models and tend to overlook state-of-the-art machine learning methods. Furthermore, the use of open-ended textual data also remains limited, with few studies leveraging modern language models and embeddings to extract meaningful patterns from qualitative responses (Hao et al., 2019).

To address these challenges, this thesis applies advanced machine learning models to a large dataset of approximately 13,000 gamers, combining psychological assessments with self-reported gaming behaviors. The study further explores how different feature representations affect predictive performance and applies interpretability techniques to identify the most influential predictors of mental health risk. Based on these goals, the following research questions were formulated:

> *To what extent can machine learning models predict mental health risk levels among video gamers based on self-reported gaming behaviors and psychological assessments, and how does the choice of feature representation (manual categorization vs. sentence embeddings) affect predictive performance?*

To address the main research question, three sub-questions were formulated:

RQ1 *How do traditional models (Logistic Regression, Random Forest, SVM) compare to advanced ensemble models (XGBoost, CatBoost, LightGBM) and deep learning architectures (TabNet) in predicting mental health risk levels?*

RQ2 *What are the most influential features for predicting mental health risk categories, as identified through SHAP analysis and feature selection techniques?*

RQ3 *Which model provides the best balance between Precision and Recall specifically for identifying individuals at high mental health risk, based on error analysis?*

3 RELATED WORK

3.1 *Mental Health and Life Satisfaction*

Mental health is a major public concern worldwide, motivating the search for proactive methods to identify individuals at risk. Traditionally, mental health has been assessed through clinical interviews, psychological

questionnaires, and observational methods (Abidogun, 2023). Despite increasing awareness, mental health disorders such as depression, anxiety and stress-related conditions remain pervasive and under-diagnosed, especially in younger populations and digitally connected communities (Kamphaus et al., 2014).

One of the key constructs in mental health research is life satisfaction, which serves as a cognitive component of subjective well-being and an indicator of psychological health (Lingan-Huaman et al., 2024; Pavot & Diener, 2008). The Satisfaction with Life Scale (SWLS) offers a validated instrument to measure people's cognitive evaluations of their life (Diener et al., 1985). It consists of five self-report items and captures the individual's subjective appraisal of their life in general. Numerous studies have demonstrated that low life satisfaction is strongly associated with anxiety, depression, and other mental health difficulties (Fergusson et al., 2015; Lombardo et al., 2018; H. Nguyen & Byeon, 2024; Shimada et al., 2020), underscoring its value as an early warning signal for psychological distress. For example, Lombardo et al., 2018 found that participants reporting low SWLS scores had significantly higher levels of depressive symptoms, while Shimada et al., 2020 further showed that diminished life satisfaction was linked to increased risk of subclinical depression in older adults.

However, traditional assessments such as SWLS require active participation and self-awareness, which can limit their effectiveness in detecting early signs of psychological distress. Emerging digital environments provide an opportunity to monitor and assess mental health indirectly. Behavioral data from social media platforms, mobile applications, and online gaming environments offer valuable real-time indicators of psychological well-being (Twenge & Farley, 2020).

3.2 *Mental Health in Gaming Communities*

The relationship between video gaming and mental health is complex, influenced by individual motivations, the social context of gaming, and engagement patterns (Birk et al., 2019; Bowman et al., 2022). Some studies suggest that excessive gaming, particularly when driven by escapism or competitive motivations, is associated with anxiety, depression, and lower life satisfaction (Finserås et al., 2022; Wei et al., 2012). Gamers classified as having problematic or addictive gaming habits often score significantly lower on the Satisfaction with Life Scale (SWLS) than non-problematic gamers, aligning with findings that low life satisfaction is associated with higher risk of Internet Gaming Disorder (IGD) and related negative outcomes (Kuss et al., 2012; Marino et al., 2020).

Johannes et al., 2021 conducted a comprehensive study, using data from Electronic Arts and Nintendo to analyze real-world gaming behavior. Contrary to concerns that excessive gaming negatively impacts mental health, they found a small but significant positive correlation between actual gameplay time and well-being. These findings suggest that psychological motives for gaming must be also considered.

Bowman et al., 2022 further explored the social dimension of gaming and its role in psychological recovery and well-being. Their study emphasized that social gaming experiences, such as multiplayer interactions, game streamings, or virtual environments, can contribute to psychological well-being and personal fulfillment. These experiences can foster feelings of social connectivity, reduce loneliness, and support emotional regulation.

At the same time, not all social gaming experiences yield positive outcomes. Sauter et al., 2021 illustrated that the underlying motivations for gaming play a crucial role in predicting mental health outcomes. Especially gamers motivated by escapism or achievement tended to report higher psychological distress, while those engaging in gaming for social connection and enjoyment experienced greater well-being.

While these findings illustrate the complex relationship between gaming behavior and mental health, they also highlight the need for more proactive approaches to identifying individuals at risk. This necessity has increasingly turned attention towards the use of machine learning techniques for mental prediction and screening.

3.3 *Machine Learning in Mental Health Prediction and Gaming Populations*

Machine learning (ML) and deep learning (DL) methods have seen rapid advancements in mental health prediction, offering novel opportunities for early detection and interventions strategies. Traditional assessment methods, although effective, are often resource-intensive and impractical for large-scale deployment.

Machine Learning in General Mental Health Prediction: In the broader mental health domain, recent studies have demonstrated the effectiveness of ML models in predicting various health conditions. Traditional models such as Logistic Regression, Support Vector Machines (SVM), K-nearest neighbors (KNN), and Decision Trees have been frequently applied due to their simplicity and interpretability. Mutalib et al. (2021) compared various algorithms to predict stress, anxiety, and depression using survey data. For stress prediction, Decision Tree achieved the highest accuracy of 84.44%, followed closely by Logistic Regression and SVM with accuracies of 82.96% and 82.22% respectively. For depression prediction, the SVM model produced the highest accuracy of 88.15%, while for anxiety predic-

tion, Logistic Regression yielded the highest accuracy at 71.85%. These results suggest that even linear models like Logistic Regression and SVM are capable of achieving high performance.

Building on these findings, Tate et al. (2020) applied a variety of machine techniques, including Random Forest, SVM, XGBoost, and Neural Networks to predict adolescent mental health outcomes using the Strengths and Difficulties Questionnaire. Their results showed that while Random Forest achieved the highest AUC (0.739), the rest of the models provided competitive performance .

Despite their success, traditional models often struggle with complex, high-dimensional datasets and non-linear relationships among psychological variables. Recent advancements have introduced ensemble methods such as XGBoost, LightGBM, and CatBoost, which have demonstrated superior predictive performance in clinical mental health contexts. Gupta et al. (2024) achieved 99.46% accuracy for postpartum depression prediction using CatBoost, while also LightGBM performed good with 98.37% , underscoring the effectiveness of ensemble learning for mental health assessments.

Additionally, deep learning models such as TabNet have recently emerged as powerful tools for tabular data, allowing automated feature extraction and achieving strong predictive performance without extensive manual preprocessing (Arik & Pfister, 2021). H. V. Nguyen and Byeon (2023) compared ensemble methods and TabNet in predicting depression using tabular data from a survey. Specifically, TabNet reached an Area under the receiver operating characteristic curve value (AUC) of 99.37% on the test set, slightly outperforming state-of-art ensemble models like XGBoost and CatBoost. These results demonstrate that deep learning can match or exceed ensemble methods on tabular data.

Machine Learning in Gaming Populations: In the gaming domain, predictive modeling of mental health outcomes is still at a relatively early stage, with studies often limited to traditional algorithms. A relevant study by Mahesh et al. (2023) used the same dataset (4.1) as the present research but focused on predicting anxiety levels using the Generalized Anxiety Disorder (GAD) scale and approached the problem through binary classification based on clinical thresholds. They utilized models like Logistic Regression, Random Forest, and Gradient Boosting with Logistic regression outperforming the others, achieving an accuracy of 85.01%.

Further extending this line of research, Hasan et al. (2024) investigated the mental health impact of mobile gaming addiction. Using a multi-class classification approach to predict mental health status, they applied KNN, SVM, Naive Bayes, and Random Forest. SVM achieved the highest accuracy of 92.75% followed by Random Forest at 85.23%. These findings reinforce

the potential of ML in addressing mental health risks associated with excessive gaming behaviors.

Another study by Vaanathy et al. (2023) focused on detecting anxiety and depression levels in video game-dependent individuals using an online survey. Applying various classifiers, including SVM, KNN, Naive Bayes, and Decision Trees, they achieved a remarkable accuracy of 93.48% for depression prediction using SVM and 91.75% for anxiety prediction using KNN. The study also highlighted the importance of feature selection techniques, which substantially improved model performance by identifying key predictors such as gaming hours, social isolation, and insomnia levels.

3.4 *Textual Feature Engineering in Mental Health Prediction*

Open-ended survey responses offer valuable qualitative insights into individuals' psychological conditions, motivations, and behaviors that are often missed by closed-ended questions. In mental health research, analyzing free-text responses can uncover hidden emotional signals, coping mechanisms, and behavior patterns that structured questions might fail to detect. Therefore, extracting relevant information from this type of textual data is essential to enhance both the predictive accuracy and the depth of mental health evaluations.

Manual categorization is a conventional method used for examining open-text responses, where human coders assign responses to predefined thematic categories (Nawaz et al., 2022). This method offers high interpretability and ensures that the analysis is closely aligned with recognized theoretical frameworks, allowing researchers to link textual content directly to concepts. For instance, in research exploring mental health among gamers, motivations were identified and manually labeled to explain psychological outcomes (Mahesh et al., 2023). However, despite its transparency, this approach is limited by human subjectivity, significant time and resource requirements, and difficulties in scaling to larger datasets (Nawaz et al., 2022).

Recent advances in natural language processing (NLP) have introduced automated techniques for processing open-ended survey responses, addressing many limitations of manual categorization. Sentence embeddings convert textual data into dense numerical vectors that capture semantic similarities and contextual nuances (Hao et al., 2019). These representations allows predictive models to utilize the rich emotional and psychological information embedded in text, which manual approaches can overlook (T. Zhang et al., 2022).

Although embeddings improve predictive performance they also introduce challenges concerning interpretability and transparency. In mental

health applications, where explainability is crucial for ethical considerations, this balance must be thoughtfully considered (Le Glaz et al., 2021). Despite these concerns, embeddings remain a promising solution for capturing complex psychological constructs from textual data and have been increasingly adopted in mental health research (Le Glaz et al., 2021; Strojny et al., 2024; T. Zhang et al., 2022).

3.5 *Research Gaps*

Despite substantial advancements in applying machine learning techniques to mental health prediction, several critical gaps persist, particularly in the context of gaming populations.

First, while ensemble learning and deep learning models have demonstrated superior predictive performance in clinical contexts, their adoption in gaming-related mental health studies remains limited. Most existing research in this domain relies on traditional classifiers such as Logistic Regression, Random Forest, and Support Vector Machines, often achieving promising results but lacking comprehensive model comparisons and rigorous evaluation frameworks (Hasan et al., 2024; Mahesh et al., 2023). This limitation prevents the establishment of reliable, state-of-the-art predictive models tailored to the unique psychological and behavioral characteristics of gaming communities.

Second, many existing studies are constrained by small datasets that limit the generalizability and robustness of their findings (Aggarwal & Goyal, 2022; Hasan et al., 2024). Small sample sizes increase the risk of overfitting and hinder the development of predictive models that can be reliably applied to broader gaming populations. There is a pressing need for research leveraging larger and more diverse datasets to enhance the validity and applicability of mental health prediction models in this field.

Third, while open-text survey responses offer valuable qualitative insights, their integration into predictive modeling remains underexplored. Existing approaches frequently rely on manual categorization, which, although interpretable, is time-consuming and lacks scalability (Nawaz et al., 2022). In contrast, modern NLP techniques such as sentence embeddings have demonstrated significant potential to improve predictive performance by capturing latent semantic information (T. Zhang et al., 2022). However, few studies have directly compared manual categorization and automated embeddings, leaving uncertainty regarding their relative effectiveness in mental health prediction tasks.

Additionally, most existing studies rely on binary classification thresholds (Mahesh et al., 2023), which oversimplify the complex spectrum of mental health risks. Few studies attempt to model this continuum through

multiclass classification, limiting the potential for nuanced mental health interventions.

In light of these gaps, this thesis aims to advance the state-of-the-art in mental health prediction for gaming populations by systematically comparing traditional and advanced machine learning models, incorporating multiclass classification, leveraging a large and diverse dataset, exploring different textual feature engineering approaches, and applying explainability techniques to enhance model transparency.

4 METHOD

This section details the methodology used to investigate the predictive performance of Machine Learning and Deep Learning models in classifying mental health risk levels among video gamers. The following subsections describe data cleaning, preprocessing, handling of missing values, and exploratory data analysis. The methodological framework consists of two parallel experiments, differing in the transformation of open-ended survey responses. One track applies manual categorization, while the other converts text responses into dense sentence embeddings. Aside from that distinction, the experiments share identical processes of data preparation, feature selection, model training, and evaluation. Seven models were implemented (Logistic Regression, Random Forest, Support Vector Machine, XGBoost, LightGBM, CatBoost, and TabNet). The dataset was split into training (70%), validation(15%) and a test set(15%). Model performance was evaluated using a combination of metrics, including F1 Macro, AUC, precision and recall. An overview of the full methodology is illustrated in Figure 1. The list of software and packages used is provided in Appendix A (7.4).

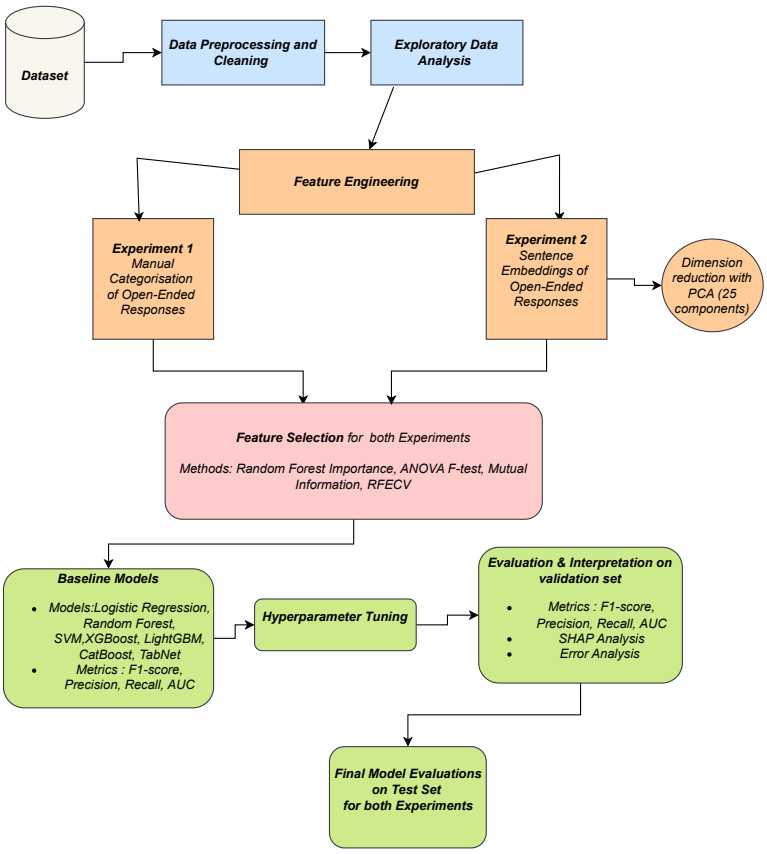

Figure 1: Overview of the Experimental Methodology Pipeline

4.1 *Dataset*

The dataset used in this study was collected through an international online survey and contains information on psychological well-being, gaming

behavior, and demographics of more than 13,000 video gamers and was obtained from the Open Science Framework (OSF) repository (Sauter & Draschkow, 2017). In total, the dataset comprises 55 features encompassing both structured and free-text information. These include standardized psychological assessment scores, such as the Satisfaction With Life Scale (SWLS), Generalized Anxiety Disorder Scale (GAD-7), and Social Phobia Inventory (SPIN), as well as demographic attributes (age, gender, education level, employment status, and country of residence). Behavioral metrics such as platform preference and playing frequency are included along with four open-ended questions related to gaming motivations, play style, monetization, and gaming rankings.

The target variable for predicting mental health risks, was derived from the total score of the SWLS and categorized into low, moderate, and high risk levels as detailed in Section 4.3.

4.2 *Data Pre-processing*

A series of preprocessing steps were undertaken to organize the dataset for future analysis. Three non-informative columns were dropped. 'Timestamp', which recorded when each participant completed the survey, did not provide any analytical value. 'Accept' merely indicated agreement with terms and conditions and was irrelevant for modeling. 'Highestleague' was found to be entirely missing and was therefore excluded. Additionally, the dataset was re-indexed using 'S. No' to clarify the order of participants and maintain a consistent reference.

4.2.1 *Handling Missing data*

Missing values appeared in both numerical and categorical features (Appendix B)(7.4).Most of them exhibited minimal missingness, with the exception of 'League' (13.76%) and 'Degree' (11.71%) followed by psychological assessment variables such as 'SPIN_T' (4.83%) and 'GADE' (4.82%). Although the extent of missing data was moderate, removing them would have substantially reduced the sample size and compromised statistical power.

Drawing from best practices in psychological and behavioral research, Multiple Imputation by Chained Equations (MICE) was applied. MICE is particularly suited for survey dataset combining numerical and categorical variables, as it takes into account the underlying data distribution and interdependencies between variables during the imputation process (Azur et al., 2011; Bruch, 2023). This approach prevents the introduction of bias often associated with simpler methods like mean or mode imputation (Azur

et al., 2011; Bruch, 2023; Woods et al., 2024). Numerical variables were imputed using Random Forest regressors within the MICE framework, while categorical features were imputed using Random Forest classifiers after ordinal encoding (Bruch, 2023). Following imputation, 'SPIN_T' was recomputed to reflect accurate totals after addressing missing item-level responses.

4.2.2 *Outliers*

Outliers detection was performed using boxplots, revealing extreme values in the 'Hours' and 'streams' variables, which represent the weekly gaming and streaming times, respectively. These outliers were handled later in Section 4.3, where the two variables were combined into a single feature, and outliers were subsequently removed during that process.

4.2.3 *Filtering*

To improve the reliability of categorical variables and reduce noise from underrepresented groups, the 'Residence_IS03' and 'Birthplace_IS03' were filtered. Only countries with at least 50 participants were retained as separate countries, while the remaining were grouped under the label 'Other'. This approach ensured more stable statistical estimates and reduced the dimensionality introduced by one-hot encoding in later modeling stages.

4.2.4 *Exploratory Data Analysis (EDA)*

Exploratory analysis of the dataset focused on understanding the distributions of key psychological assessment scores along with their relationships to demographic and behavioral features. This step provided vital insights into the psychological and behavioral tendencies of the gaming population and guided the selection of future steps.

Correlations between numerical variables and Satisfaction With Life Total Scores (SWL_T) were computed to identify potential predictors. The strongest correlations by absolute value are visualized in Figure 2. Due to its size, the full correlation matrix is available in Appendix C 7.4. A moderate negative correlation was identified between life satisfaction (SWL_T) and generalized anxiety (GAD_T)(0.40%), indicating that individuals with higher levels of anxiety are likely to report lower life satisfactions. Similarly, SWL_T showed a moderate negative correlation with social phobia symptoms (SPIN_T)(0.30%).

Satisfaction with Life (SWL_T) Distribution: Figure 3 exhibits a distribution that is close to normal with a slight left skew, indicating a concentration of participants with moderate to high life satisfaction.

Additionally, exploratory visualizations (Figures 4,5,6) were included to examine variations in life satisfaction (SWL_T) across different participant subgroups.

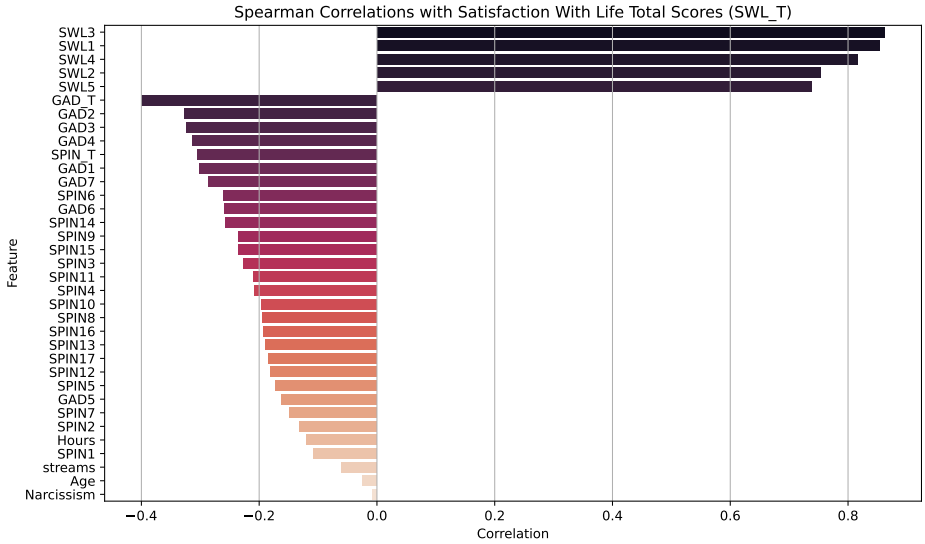

Figure 2: Spearman Correlations of numeric features with SWL_T, sorted by the absolute strength of correlation. The plot highlights features most strongly associated (positively or negatively) with SWL_T

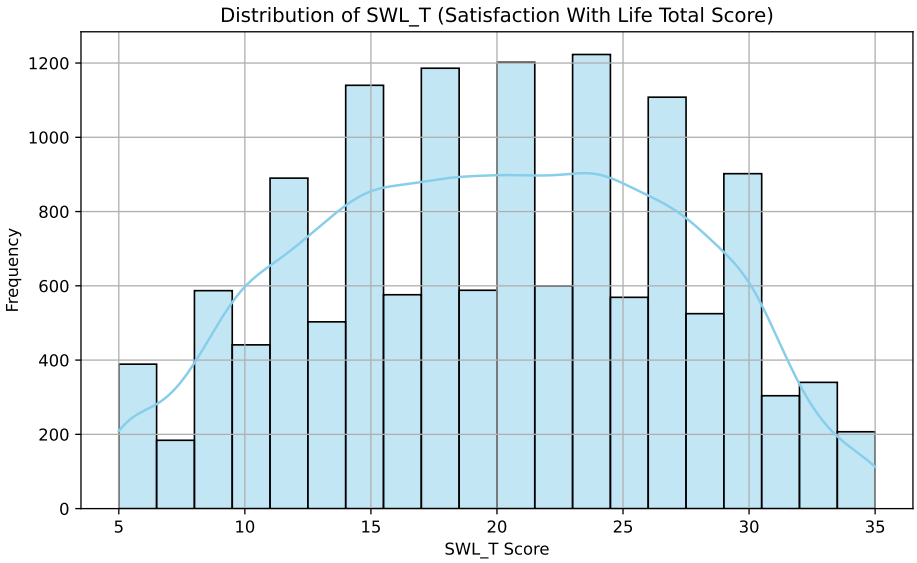

Figure 3: Distribution of SWL_T. This histogram shows the distribution of participants' total scores on the SWLS. The scores range from 5 to 35, with a left-skewed distribution.

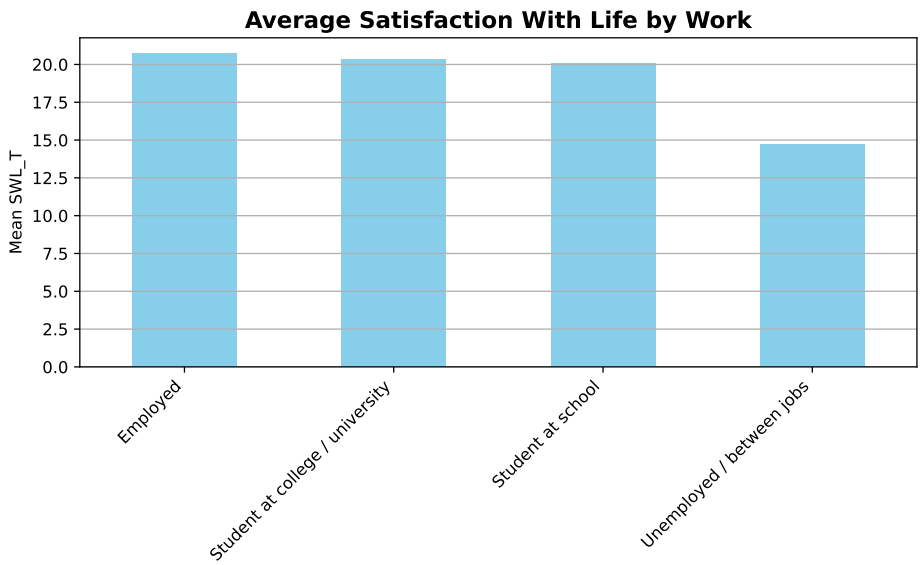

Figure 4: Average SWL_T scores by employment status. The bar chart presents the mean SWL_T scores across the employment status.

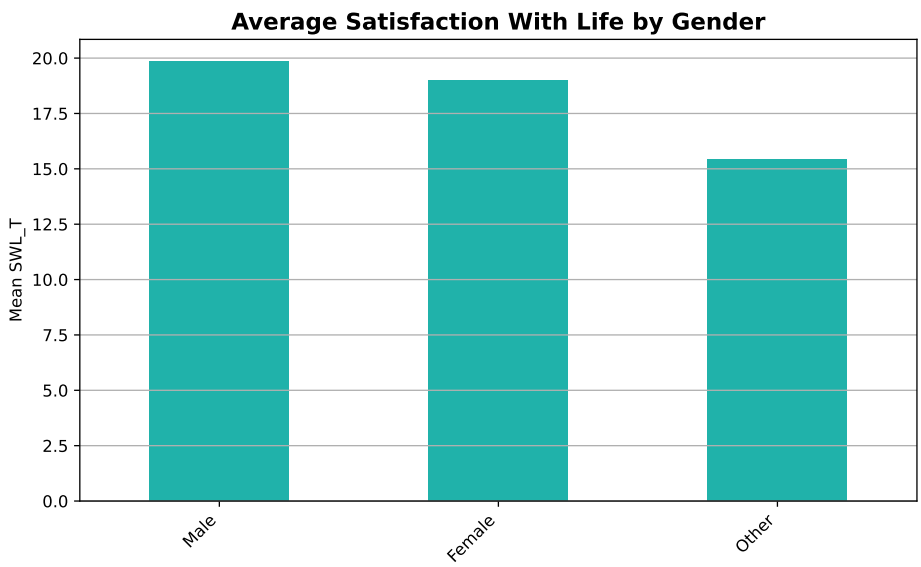

Figure 5: Average SWL_T scores by gender. The bar chart presents the mean SWL_T scores across genders.

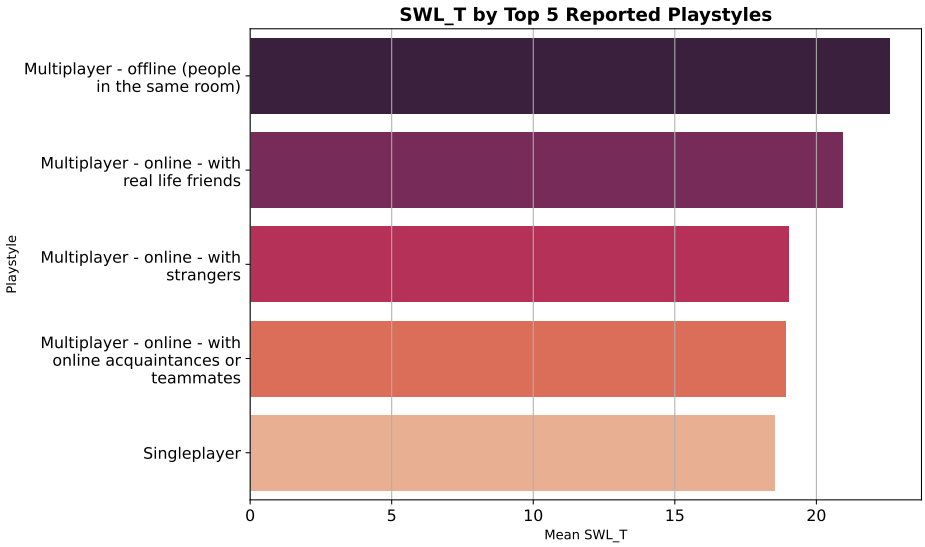

Figure 6: Average SWL_T scores by the five most frequently reported Playstyles. The chart displays the mean life satisfaction scores across these playstyles, showing differences in well-being associated with self-reported gaming motivations.

4.3 *Feature Engineering*

Feature engineering includes several key steps, enabling the transformation of raw survey data into meaningful predictors for mental health classification. This section involved the creation of the target variable, additional feature construction, and the two parallel experiments conducted to handle openended survey responses.

The features *'Hours'* (weekly gaming hours) and *'streams'* (weekly streaming hours) were combined into a new feature named *'GamingTime'*, to capture a comprehensive view of participants engagement. Outlier handling was performed during this step by removing participants who reported implausible engagement times exceeding 140 hours per week. As a result, 20 observations were dropped.

Target Creation : As observed in several studies, researchers often transform continuous psychological assessment scores into categorical classes to facilitate classification tasks and simplify the interpretation of mental health risks (Mutalib et al., 2021; H. V. Nguyen & Byeon, 2023; Sau & Bhakta, 2019). This approach enables models to identify distinct risk groups (low, moderate, high), rather than working directly with continuous outcomes, which can be harder to interpret in practical intervention settings. The target variable, *'mental_health_risk'*, was derived from the Satisfaction with Life Scale total score (SWL_T). Following established thresholds (Diener

et al., 1985), participants were first categorized into life satisfaction levels. Subsequently, these categories were linked to the corresponding levels of mental health risks based on the psychological literature that emphasized that people with low satisfaction are substantially more likely to experience symptoms related to anxiety and other psychological problems (Fergusson et al., 2015; Lombardo et al., 2018).

Table 1 summarizes the categorization scheme applied.

Table 1: Mapping of SWL_T Scores to Mental Health Risk Levels

SWL_T Range	Life Satisfaction Category	Mental Health Risk
31 – 35	Extremely Satisfied	Low Risk
26 – 30	Satisfied	Low Risk
21 – 25	Slightly Satisfied	Low Risk
20	Neutral	Moderate Risk
15 – 19	Slightly Dissatisfied	Moderate Risk
10 – 14	Dissatisfied	High Risk
5 – 9	Extremely Dissatisfied	High Risk

The dataset included four survey questions related to gaming motivations: *'whyplay'*, *'Playstyle'*, *'earnings'*, and *'League'*. While these were primarily multiple-choice questions with predefined response categories, participants were also given the option to provide free-text input. This introduced variability that required additional processing to standardize responses. Two parallel experiments were conducted to evaluate the impact of different transformation techniques for these features.

Experiment 1 - Manual Categorization: In the first experiment, responses were transformed into structured categorical variables using a rule-based approach. Predefined domain-relevant keywords were used to classify free-text responses into meaningful responses. For example, *whyplay* feature included categories such as *"having fun"*,*"improving"*,*"winning"*, *"relaxing"*,*"escaping"*. Similar mappings were applied to the other features. This approach was selected for its high interpretability and alignment with established practices in behavioral and social sciences. Manual coding of survey responses ensures direct mapping to psychologically relevant constructs and supports model explainability. Although this strategy cannot capture every subtle nuance in the text, it remains widely used in survey research (Card & Smith, 2015).

Experiment 2 - Sentence Embeddings: To overcome the limitations of manual categorization, a second experiment converted all four items into dense semantic vectors. We adopted *all-MiniLM-L6-v2* Sentence Transformer (Reimers & Gurevych, 2025), since it delivers strong performance while remaining computationally lightweight (Pearson, 2024). This model

produces dense vector representations that encode the latent semantic content of textual responses, even when responses are brief or highly variable. Sentence embeddings have been demonstrated to outperform traditional word-level embeddings in encoding subjective survey data and improving predictive validity (Fang et al., 2022). Each response was encoded into a 384-dimensional dense vector, resulting in a combined embedding matrix of shape (13,444, 1536).

To mitigate the curse of dimensionality and prevent overfitting, Principal Component Analysis (PCA) (Ma*f*ákiewicz & Ratajczak, 1993; van der Meer et al., 2024) was applied. The decision to retain 25 components, preserving over 90% of the variance, aligns with the findings by G. Zhang et al., 2024, who evaluated dimensionality reduction techniques for sentence embeddings and confirmed that PCA offers a strong balance between information retention, semantic structure preservation, and computational efficiency. This approach ensured that key semantic patterns are maintained while minimizing overfitting risks.

Both experiments followed identical preprocessing, feature selection, modeling, and evaluation pipelines.

4.4 *Stratified Sampling, Encoding, and Scaling*

Before splitting and scaling the data, we removed several columns that were either directly used to construct the target variable (*SWL_T*, *SWL1*, *SWL2*, *SWL3*, *SWL_category*) or replaced by transformed versions (*whyplay, Playstyle, earnings, League*). This ensured that no information leakage or redundancy influenced the model training process.

As shown in Figure 7, the dataset does not exhibit a severe class imbalance. Approximately 47.7% of participants fall into the *Low Risk* category, while *Moderate Risk* and *High Risk* categories are almost equally represented, each accounting for roughly 26.3% of the sample.

The dataset was split into 70% for training, 15% for validation, and 15% for testing. All model development steps were strictly confined to the training set. Within this subset, stratified 5-fold cross-validation was employed for baseline performance estimation and hyperparameter tuning. The validation set (15%) was used as an out-of-fold evaluation set to compare the performance of the tuned pipelines and to conduct SHAP explainability and error analysis. The test set (15%) remained completely untouched until the final evaluation phase to ensure an unbiased assessment of the selected model's performance.

Two parallel versions of dataset (Table 2) were created to accommodate the requirements of different algorithms. For models sensitive to feature magnitudes -such as Logistic Regression, Random Forest, Support Vector

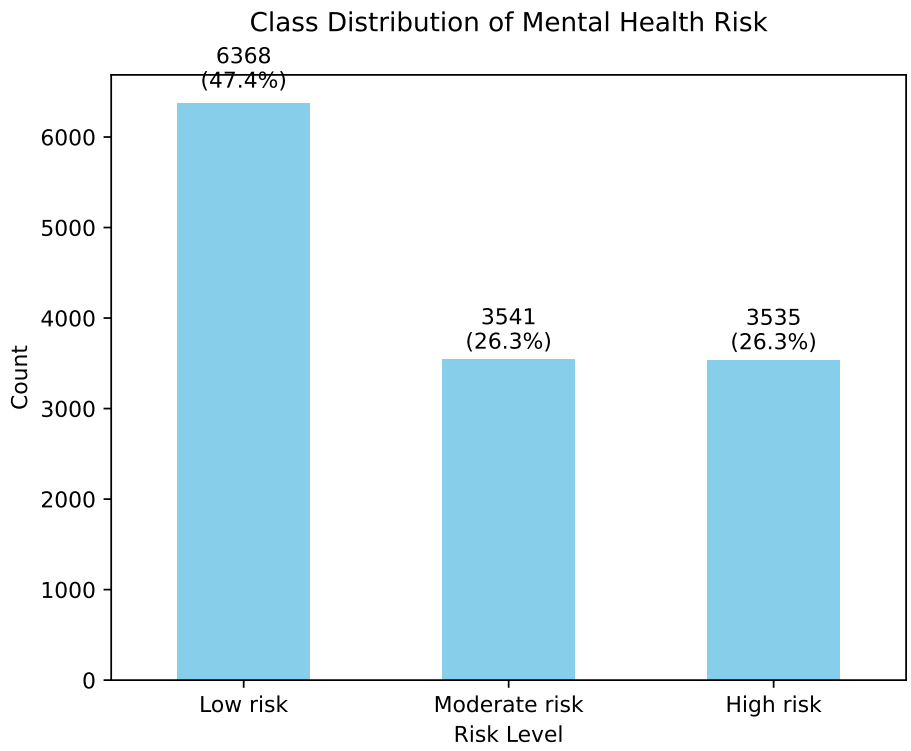

Figure 7: Class distribution of the target variable *mental_health_risk*, showing the number and proportion of participants classified into low, moderate, and high risk groups based on Satisfaction With Life Total scores (SWL_T).

Machine, XGBoost, LightGBM- categorical features were one-hot encoded, and numerical features were standardized using z-score normalization via *StandardScaler*. In contrast, for CatBoost, the dataset was retained in its raw form without encoding, since it is capable of handling raw categorical features directly.

Table 2: Dataset Splits for Preprocessed and Raw Versions

Dataset Version	**Training Set**	**Validation Set**	**Test Set**
Encoded + Scaled	9,410	2,017	2,017
Raw (Unprocessed)	9,410	2,017	2,017

4.5 *Feature Selection*

We employed multiple methods for feature importance evaluation and selection, each offering complementary insights into the relevance of the

features: Random Forest feature importance, ANOVA F-test, Mutual Information, and Recursive Feature Evaluation with cross-validation.

Random Forest's built-in *feature_importances_* attribute calculates importance based of the mean decrease in impurity across decision trees (Breiman, 2001). This methods is particularly effective for ensemble models and provides a reliable measure of a feature's predictive strength. Bar plots of the top 40 most important features for both experiments were generated to visually present the results (Figure 9, Figure 8). Notably, *GamingTime*, *GAD_T*, and *SPIN_T* consistently appeared as top predictors across both experiments, confirming their strong predictive relevance.

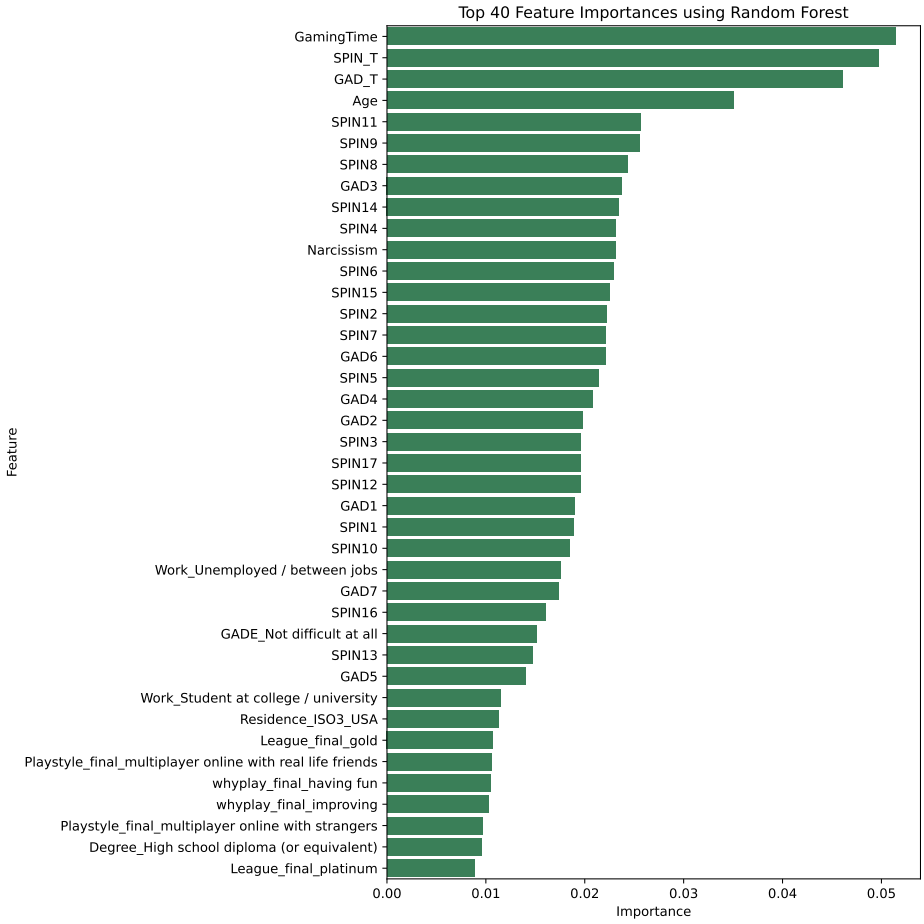

Figure 8: Random Forest feature importance ranking for Experiment 1. The top 40 features are displayed based on their contribution to the model's predictions, with higher values indicating greater predictive relevance.

The one-way ANOVA F-test measures how much of a numerical feature's variability can be attributed to class membership. Features whose group means differ significantly are flagged as informative and can be

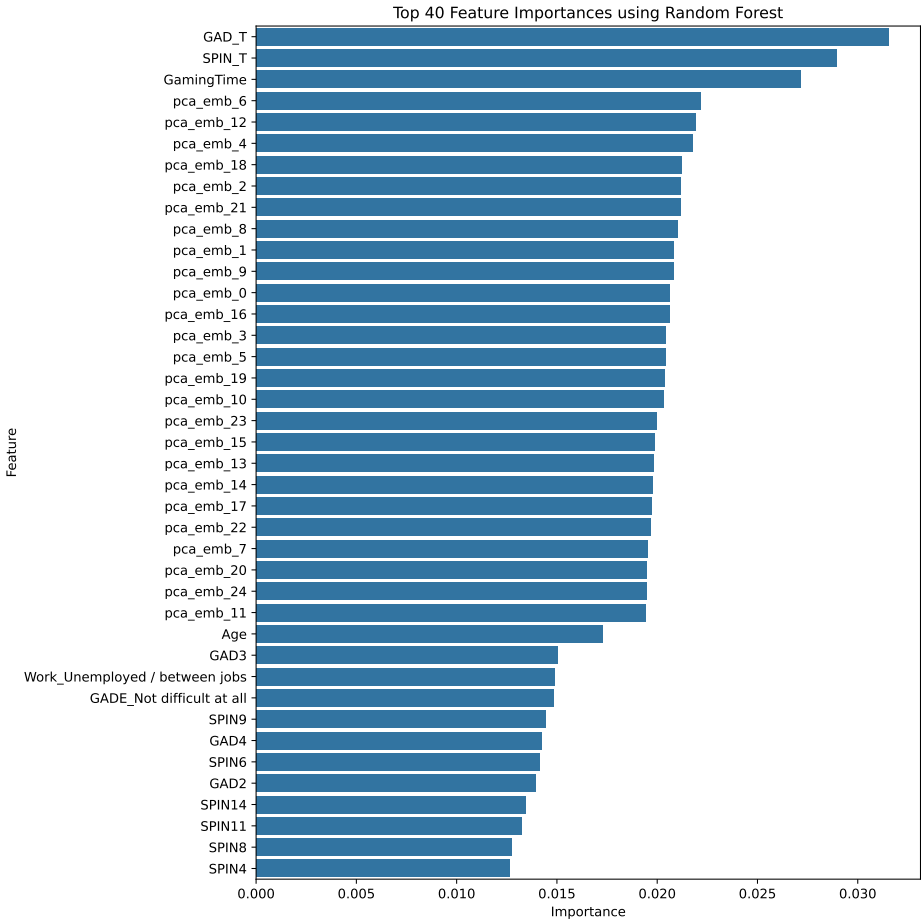

Figure 9: Top 40 Feature Importances Using Random Forest (Experiment 2-Features). The top 40 features are displayed based on their contribution to the model's predictions, with higher values indicating greater predictive relevance.

ranked by their F-statistics, making the test a straightforward filter for variables that show near-linear separation across classes (D. Chen et al., 2005). The analysis revealed that psychological indicators, along with behavioral factors and employment status, exhibited the strongest relationships. These features consistently recorded the highest F-scores and statistically significant p-values ($p < 0.001$) on both experiments. A table for the top 10 evaluated features is available in Appendix D (7.4). In parallel, Mutual Information measures the amount of information shared between each feature and the target variable, effectively capturing nonlinear relationships (Kraskov et al., 2004).

Finally, we applied RFECV with both Logistic Regression and XGBoost classifiers to iteratively remove the less relevant features (Awad & Fraihat, 2023). The selection criterion was based on maximizing the F1 Macro score through $5 - fold$ stratified cross-validation. This wrapper method balances predictive performance with feature reduction.

Despite differences in methodology, significant overlap was observed among the top-ranked features identified by all methods. This consistency reinforces the reliability of these features in predicting mental health risk. Psychological assessments such as *GAD_T*(Generalized Anxiety Disorder), *SPIN_T*(Social Phobia Inventory), as well as their individual scores, consistently emerged as the strongest predictors of mental health risk. Functional impairment *GADE* and employment status *Work* were frequently identified as important features, underscoring the influence of socio-economic and daily functioning factors on mental health. Notably, in Experiment 2, several PCA components derived from sentence embeddings also ranked highly, indicating that these representations captured meaningful information related to gaming motivations and behaviors.

These findings highlight the complex interplay between psychological conditions, socio-economic status, and behavioral patterns in shaping mental health outcomes. The added value of text-based features further suggest that individuals' self-reported gaming experiences provide valuable contextual information.

Based on the results of feature selection and importance analyses, five distinct feature sets were generated to investigate how different levels of feature dimensionality influence model performance.

Features that appeared in at least three selection methods were combined to form the *Consensus* set. This intersection -based approach ensures the retention of features that demonstrated consistent relevance across diverse selection techniques, thereby enhancing the robustness and generalizability of the final models. Additionally, this method helps mitigate the risk of overfitting by excluding features that only show importance in isolated methods.

Table 3 and Table 4 provide an overview of the resulting feature sets for both experiments.

Table 3: Overview of Final Feature Sets Used for Modeling - Experiment 1 - Manual Categorization

Feature Set	**Number of Features**
Full Dataset	106
RFECV Logistic Regression	105
RFECV XGBoost	88
Consensus ($\geq$ 3 Methods)	64
Top-20 Random Forest	20

Table 4: Overview of Final Feature Sets Used for Modeling - Experiment 2 - Embeddings

Feature Set	**Number of Features**
Full Dataset	183
RFECV Logistic Regression	166
RFECV XGBoost	109
Consensus ($\geq$ 3 Methods)	70
Top-20 Random Forest	20

4.6 *Algorithms*

4.6.1 *Baseline Models*

Three baseline models were employed: *Logistic Regression*, *Random Forest*, and *Support Vector Machine*.

The Logistic Regression model was selected for its simplicity, interpretability and effectiveness in handling linearly separable data (David W. Hosmer Jr., 2013). Despite its linear nature, it remains a strong baseline for multiclass classification problems when appropriate regularization is applied to prevent overfitting.

Random Forest is an ensemble learning algorithm that constructs multiple decision trees and aggregates their prediction to improve classification accuracy and robustness (Breiman, 2001). Using the technique of *bagging*, RF randomly samples subsets of the data with replacement to train each individual tree. Each tree independently votes for a class label, and the final prediction is determined by majority voting. This method effectively

reduces overfitting by averaging across multiple models, thereby improving generalization performance.

Support Vector Machines (SVMs) are powerful classifiers designed to find the optimal hyperplane that maximizes the margin between classes (Cortes & Vapnik, 1995). SVM is particularly effective in high-dimensional spaces and can model complex, non-linear relationships using kernel functions.

4.6.2 *Additional Models*

In addition to the baseline models, four advanced algorithms were employed to explore potential improvements in model performance and robustness: *XGBoost*, *LightGBM*, *CatBoost*, and *TabNet*. These models were selected based on their proven effectiveness in handling complex, high-dimensional data and their capacity to capture non-linear relationships, which are common on psychological and behavioral datasets.

Gradient boosting algorithms (*XGBoost*, *LightGBM*, *CatBoost*) have demonstrated strong performance in predictive modeling tasks, particularly when working with structured and tabular data.

XGBoost (eXtreme Gradient Boosting), is a scalable and highly optimized implementation of gradient boosting decision trees, introduced by T. Chen and Guestrin, 2016. It utilizes a sequential learning approach where each weak learner corrects the errors of its predecessors, enhancing overall model performance.

LightGBM (Ke et al., 2017) introduces innovations such as histogram-based decision tree learning and leaf-wise tree growth, which significantly improve computational efficiency and reduce memory consumption. These characteristics make it particularly suitable for large datasets and fast experimentation.

CatBoost (Prokhorenkova et al., 2018) focuses on the efficient handling of categorical variables and employs ordered boosting to reduce prediction bias. Its advanced processing of categorical features minimizes the need for extensive preprocessing and makes it highly effective in scenarios with mixed data types.

TabNet is a deep learning architecture tailored for tabular data, introduced by Arik and Pfister (2021). Leveraging sequential attention mechanisms, TabNet learns feature representations dynamically at each decision step, allowing it to focus on the most relevant features while suppressing less informative ones. This interpretability, combined with its capacity to model non-linear patterns, makes it a compelling choice for exploring whether deep learning architectures can outperform traditional machine learning models in predicting mental health risk. Additionally, the attention-based explanations provided by TabNet facilitate understand-

ing the decision-making process, which is critical for research in sensitive domains such as mental health.

All models were evaluated using a consistent *5-fold* stratified cross-validation framework across the five feature sets to ensure methodological rigor and comparability of results. Hyperparameter optimization was conducted to improve predictive performance and model generalization, with further details provided in the Section 4.6.3.

4.6.3 *Hyperparameter Tuning*

Although multiple feature sets were generated and considered during the initial phase of model development, the RFECV_XGB feature set was selected for all subsequent hyperparameter tuning and final model evaluations. This decision was informed by preliminary modeling runs, which indicated that RFECV_XGB provided a strong balance between predictive performance and feature dimensionality. Additionally, given the time constraints of the project, focusing on a single, feature set ensured the feasibility of conducting extensive hyperparameter optimization without compromising the depth of the analysis.

To enhance the predictive performance of the models, hyperparameter tuning was conducted using a combination of *GridSearch* (for Logistic Regression,RF, SVM) and *Randomized Search* (for XGBoost, LightGBM, CatBoost, TabNet) strategies, depending on the complexity of the model. A *5-fold* stratified cross-validation technique was utilized in all experiments to ensure robustness and mitigate the risk of overfitting. The primary evaluation metric guiding the tuning process was *Macro F1 score*, chosen for its balanced approach to class imbalances and overall predictive effectiveness.

Tuning Logistic Regression: The key hyperparameters of the logistic regression model (summarized in Table 5) were tuned with guidance from earlier work on mental health prediction (Ding et al., 2025; Y. Zhang et al., 2025). The tuning process focused on the regularization strength (C), penalty type ($L1$, $L2$), and solver selection. These hyperparameters affect the model's complexity and regularization behavior, which are critical for multiclass classification tasks.

Table 5: Hyperparameter Tuning for Logistic Regression

Hyperparameter	**Values**
C	{0.01, 0.1, 1, 10, 100}
multi-class Strategy	{ovr, multinomial}
$class_weight$	{None, balanced}
$solver$	$lbfgs$
max_iter	1000
$randomstate$	42

Tuning Random Forest: The tuning process (Table 6) mainly focused on controlling model variance and bias through the number of trees (*n_estimators*), maximum tree depth, minimum samples per leaf, and split criteria. This approach is consistent with Breiman's recommendations for enhancing ensemble model stability and predictive power (Breiman, 2001) and also follows the implementations of Y. Zhang et al., 2025.

Table 6: Hyperparameter Tuning for Random Forest

Hyperparameter	**Values**
n_estimators	{50, 100, 200}
max_depth	{10, 20, 30, None}
$min_samples_leaf$	{1, 3, 5, 10}
splitting criterion	{gini, entropy}
randomstate	42

Tuning SVM: Given the model's sensitivity to parameter settings such as C and *gamma*, we emphasized on regularization parameter C, kernel type, kernel coefficient *gamma*, and the use of class weighting to address potential class imbalances, as also introduced by Bergstra et al., 2011; Cortes and Vapnik, 1995 (Table 7).

Table 7: Hyperparameter Tuning for Support Vector Machine (SVM)

Hyperparameter	**Values**
C	{0.1, 1, 10}
kernel	{linear, rbf}
gamma	{scale, auto}
classweight	{None, balanced}
Probability Estimates	True
randomstate	42)

Tuning XGBoost and LightGBM: Primary hyperparameters (Tables 8 and 9) include the learning rate, maximum tree depth, number of estimators, and regularization parameters (*lambda*, *gamma*). Both models utilize gradient boosting methods, where managing these parameters can prevent overfitting (T. Chen & Guestrin, 2016; Ke et al., 2017).

Table 8: Hyperparameter Tuning for XGBoost

Hyperparameter	**Values**
n_estimators	{300, 350, 400}
max_depth	{8, 10, 12}
learning_rate	{0.03, 0.05, 0.07}
gamma	{0.05, 0.1, 0.2}
colsample_bytree	{0.7, 0.8, 0.9}
subsample	{0.7, 0.8, 0.9}
reg_lambda	{5, 10, 15}
randomstate	42

Table 9: Hyperparameter Tuning for LightGBM

Hyperparameter	**Values**
n_estimators	{100, 200, 300}
learning_rate	{0.01, 0.05, 0.1}
max_depth	{5, 7, 10}
num_leaves	{15, 31, 63}
min_child_samples	{10, 20, 30}
subsample	{0.7, 0.8, 0.9}
colsample_bytree	{0.7, 0.8, 0.9}
reg_lambda	{0, 1, 5, 10}
randomstate	42

Tuning CatBoost: The tuning process concentrated on essential hyperparameters, including the number of boosting iterations, tree depth, learning rate, and regularization parameters. Additionally, parameters such as *bagging temperature* and *random strength* were optimized to manage overfitting and improve the robustness of the model (Prokhorenkova et al., 2018). Class weights were utilized, calculated using the balanced class distribution from the training dataset.

Table 10: Hyperparameter Tuning for CatBoost

Hyperparameter	**Values**
iterations	{200, 300, 400}
depth	{4, 6, 8, 10}
learning_rate	{0.01, 0.03, 0.05}
l2_leaf_reg	{1, 3, 5, 7, 10}
bagging_temperature	{0, 0.5, 1}
random_strength	{1, 5, 10}
randomstate	42

Tuning TabNet: For TabNet (Table 11), the search concentrated on architecture size, depth, and regularization, followed the recommendations of Arik and Pfister, 2021. The width of both the decision and attention embeddings (*n_d*, *n_a*) was varied alongside the number of sequential decision steps, while sparsity and feature-mask smoothness were governed by the *lambda* and *gamma*. Learning-rate schedules and optimizer settings were also probed: several step sizes were paired with the Adam optimizer, with optional step-decay schedulers to curb over-fitting.

Table 11: Hyperparameter Tuning for TabNet

Hyperparameter	**Values**
n_d	{8, 16, 24, 32}
n_a	{8, 16, 24, 32}
n_steps	{3, 5, 7}
gamma	{1.0, 1.3, 1.5}
lambda_sparse	{1e-5, 1e-4, 1e-3}
momentum	{0.02, 0.05, 0.1}
optimizer	Adam
$learning_rate$	{0.005, 0.01, 0.02}
scheduler	{StepLR, None}
masktype	{Sparsemax, Entmax}
randomstate	42

4.7 *Experimental Set-up*

Both experiments followed a standardized pipeline to ensure comparability and reproducibility. After hyperparameter tuning with stratified five-fold cross-validation, the best configuration of each model was assessed on an unseen validation fold, using the macro-averaged F1 score as the primary

metric. Most analyses relied on the RFECV-XGB feature subset; however, CatBoost was additionally evaluated on the full, unprocessed feature set to exploit its native treatment of categorical variables.

Model behavior was interpreted with SHAP (SHapley Additive exPlanations) (Lundberg & Lee, 2017). Global summaries highlighted the most influential predictors, while local explanations clarified the feature contributions driving individual classifications. Particular attention was devoted to the Moderate-risk class, which remained the hardest to predict accurately.

Finally, confusion matrices, class-specific metrics, and SHAP comparisons between correctly and incorrectly labeled cases were examined to pinpoint systematic error patterns. Together, these interpretability and error analyses illuminate both the strengths and the limitations of the proposed models

4.8 *Model Evaluation*

The performance of the trained models was evaluated using various metrics to provide a comprehensive and balanced assessment. The primarily evaluations metric was the $Macro\ F1 - score$, chosen for its ability to equally account for each class. This choice was particularly important in the context of mental health classification, where the Moderate risk group was slightly underrepresented and more challenging to classify.

Additionally, $Precision$ and $Recall$ with $macro - averaging$ were reported to provide further insight into the model's ability to correctly identify each mental health risk category. These metrics helped assess the trade-off between false positives and false negatives which is critical in a mental health-related predictive task.

The $ROC - AUC$ (One-vs-Rest) metric was also computed to assess the models' capability to differentiate between classes using predicted probabilities. Although *accuracy* was included for completeness, it was not emphasized in the evaluation process, as it can be misleading in multiclass classification tasks where class imbalances, even if moderate, can distort performance interpretations.

Despite the dataset being moderately balanced, the decision to prioritize macro-averaged metrics ensured that performance assessments remained sensitive to the needs of all classes, rather than favoring the majority class. This approach aligns with the goal of developing models that perform robustly across all mental health risk levels.

5 RESULTS

5.1 *Baseline Models*

Table 12 and Table 13 presents the baseline results using 5-fold stratified cross-validation for the manual categorization and the embedding-based experiments across all models and feature sets. Performance was assessed through F1-score, Precision, Recall and AUC (One-vs-Rest).

Overall for **Experiment 1** the models exhibited moderate classification performance, with F1 Macro scores ranging between 0.40 and 0.48 across most feature sets. LightGBM and CatBoost consistently achieved the highest F1 scores, with CatBoost reaching a peak value of 0.4799. TabNet, despite its advanced architecture, did not surpass the tree-based models, reaching a value of 0.4326.

In **Experiment 2**, baseline performance remained modest across models, with F1-macro scores ranging from 0.42 to 0.48. As in the Experiment 1, CatBoost achieved the highest performance, reaching a maximum F1 score of 0.4793 on the Raw feature set (without encoding or scaling). TabNet again demonstrated lower performance, achieving a maximum F1-score of 0.4309.

Notably, for both experiments, performance differences across feature sets were marginal for all models, suggesting that feature selection had limited impact on predictive performance. Consequently, for efficiency and consistency, subsequent hyperparameter tuning and final evaluations were conducted primarily using the *RFECV_XGB* feature set. CatBoost was an exception, evaluated an the *Raw* feature set due to its native handling of categorical features and slightly better baseline performance.

Overall, while the baseline results provide a starting point, the modest performance across models highlights the need for further optimization through hyperparameter tuning.

Table 12: Experiment 1 (Manual Categorization): Baseline Model Evaluation on Training Set (5-Fold Cross-Validation, Mean Values)

Model	Feature Set	F1 Score	Precision	Recall	AUC	Accuracy
Logistic Regression	Full	0.4582	0.4936	0.4873	0.6996	0.5642
	RFECV_LogReg	0.4593	0.4958	0.4880	0.6998	0.5646
	RFECV_XGB	**0.4605**	0.4980	0.4894	0.7002	0.5659
	Consensus	0.4576	0.4997	0.4886	0.7016	0.5666
	Top20_rf	0.4088	0.4718	0.4599	0.6755	0.5471
Random Forest	Full	0.4472	0.4830	0.4759	0.6838	0.5533
	RFECV_LogReg	0.4537	0.4947	0.4794	0.6885	0.5578
	RFECV_XGB	0.4531	0.4950	0.4789	0.6853	0.5570
	Consensus	**0.4567**	0.4921	0.4793	0.6875	0.5553
	Top20_RF	0.4402	0.4677	0.4608	0.6578	0.5352
SVM	Full	0.4398	0.4882	0.4736	0.6866	0.5571
	RFECV_LogReg	0.4401	0.4884	0.4739	0.6868	0.5573
	RFECV_XGB	0.4405	0.4896	0.4739	0.6865	0.5574
	Consensus	**0.4420**	0.4905	0.4747	0.6857	0.5576
	Top20_rf	0.4129	0.4797	0.4559	0.6475	0.5443
XGBoost	Full	0.4689	0.4809	0.4771	0.6750	0.5392
	RFECV_LogReg	0.4715	0.4885	0.4838	0.6750	0.5451
	RFECV_XGB	**0.4760**	0.4822	0.4795	0.6738	0.5403
	Consensus	0.4714	0.4823	0.4790	0.6738	0.5399
	Top20_rf	0.4442	0.4578	0.4525	0.6406	0.5157
LightGBM	Full	0.4737	0.4939	0.4882	0.6893	0.5560
	RFECV_LogReg	0.4699	0.4909	0.4853	0.6895	0.5550
	RFECV_XGB	**0.4751**	0.4979	0.4899	0.6916	0.5584
	Consensus	0.4736	0.4955	0.4874	0.6940	0.5558
	Top20_rf	0.4430	0.4676	0.4604	0.6590	0.5316
CatBoost	Full	0.4779	0.4948	0.4897	0.6906	0.5531
	RFECV_LogReg	0.4780	0.4977	0.4899	0.6923	0.5540
	RFECV_XGB	**0.4799**	0.4943	0.4881	0.6877	0.5519
	Consensus	0.4749	0.4893	0.4852	0.6867	0.5479
	Top20_rf	0.4477	0.4625	0.4580	0.6488	0.5225
	Raw	0.4741	0.4939	0.4898	0.6942	0.5575
TabNet	Full_RULE	0.4318	0.4767	0.4852	0.6879	0.5638
	RFECV_LogReg	0.4314	0.4659	0.4839	0.6847	0.5625
	RFECV_XGB	0.4207	0.4769	0.4840	0.6858	0.5632
	Consensus	0.4211	0.5112	0.4788	0.6845	0.5625
	Top20_rf	0.4052	0.4257	0.4656	0.6663	0.5485
	Raw	**0.4326**	0.4878	0.4750	0.6697	0.5496

Table 13: Experiment 2 (Sentence Embeddings): Baseline Model Evaluation on Training Set (5-Fold Cross-Validation, Mean Value)

Model	Feature Set	F1 Score	Precision	Recall	AUC	Accuracy
Logistic Regression	Full	0.4606	0.4926	0.4869	0.6962	0.5624
	RFECV_LogReg	0.4562	0.4864	0.4841	0.6980	0.5607
	RFECV_XGB	**0.4621**	0.4984	0.4894	0.6991	0.5657
	Consensus	0.4587	0.5021	0.4901	0.6985	0.5675
	Top20_rf	0.3972	0.4463	0.4528	0.6694	0.5419
Random Forest	Full	0.4372	0.4757	0.4643	0.6709	0.5440
	RFECV_LogReg	0.4490	0.4909	0.4747	0.6773	0.5533
	RFECV_XGB	0.4434	0.4729	0.4688	0.6732	0.5460
	Consensus	**0.4418**	0.4692	0.4672	0.6729	0.5425
	Top20_rf	0.4226	0.4294	0.4304	0.6206	0.4889
SVM	Full	0.4346	0.4855	0.4711	0.6805	0.5562
	RFECV_LogReg	0.4345	0.4773	0.4696	0.6831	0.5538
	RFECV_XGB	**0.4351**	0.4858	0.4716	0.6806	0.5566
	Consensus	0.4331	0.4784	0.4686	0.6801	0.5533
	Top20_rf	0.3983	0.4534	0.4406	0.6389	0.5332
XGBoost	Full	0.4652	0.4767	0.4730	0.6741	0.5352
	RFECV_LogReg	0.4688	0.4802	0.4761	0.6715	0.5377
	RFECV_XGB	**0.4693**	0.4815	0.4771	0.6710	0.5392
	Consensus	0.4709	0.4823	0.4772	0.6721	0.5371
	Top20_rf	0.4338	0.4419	0.4419	0.6322	0.5034
LightGBM	**Full**	**0.4689**	0.4862	0.4827	0.6832	0.5506
	RFECV_LogReg	0.4677	0.4871	0.4815	0.6843	0.5491
	RFECV_XGB	0.4649	0.4832	0.4807	0.6837	0.5496
	Consensus	0.4658	0.4848	0.4797	0.6829	0.5486
	Top20_rf	0.4316	0.4453	0.4471	0.6448	0.5165
CatBoost	Full	0.4758	0.4968	0.4900	0.6870	0.5548
	RFECV_LogReg	0.4767	0.4950	0.4888	0.6847	0.5538
	RFECV_XGB	0.4775	0.4946	0.4883	0.6837	0.5528
	Consensus	0.4779	0.4932	0.4881	0.6830	0.5506
	Top20_rf	0.4345	0.4444	0.4451	0.6377	0.5082
	Raw	**0.4793**	0.4932	0.4878	0.6834	0.5521
TabNet	Full	0.4246	0.4563	0.4792	0.6873	0.5607
	RFECV_LogReg	0.4262	0.4540	0.4798	0.6798	0.5607
	RFECV_XGB	**0.4309**	0.5126	0.4866	0.6848	0.5606
	Consensus	0.4269	0.4296	0.4834	0.6830	0.5625
	Top20_rf	0.3954	0.5175	0.4593	0.6643	0.5438
	Raw	0.4120	0.5118	0.4690	0.6733	0.5550

5.2 *Hyperparameters tuning*

A comprehensive hyperparameter tuning procedure was conducted to optimize model performance across both experiments. The tuning focused on maximizing the F1 Macro score using 5-fold-stratified-cross-validation.

For both **Experiment 1** and **Experiment 2** hyperparameter optimization was carried out using Grid Search for simpler models (Logistic Regression, RF, and SVM), allowing for an exhaustive search over specified parameter grids. For more complex models like XGBoost, LightGBM, CatBoost, TabNet, Randomized Search was employed to efficiently explore larger hyperparameter spaces within a reasonable computational budget.

The optimal hyperparameters identified during this process are summarized in Tables 14 and 15, and were used in the model evaluations.

Table 14: Experiment 1: Optimal Hyperparameters for Each Model

Model	**Best Hyperparameters**
Logistic Regression	{'C': 0.01, 'class_weight': 'balanced', 'multi_class': 'multinomial'}
Random Forest	{'criterion': 'gini', 'max_depth': 20, 'min_samples_leaf': 1, 'n_estimators': 100}
SVM	{'C': 1, 'class_weight': 'balanced', 'gamma': 'scale', 'kernel': 'linear'}
XGBoost	{'subsample': 0.7, 'reg_lambda': 5, 'n_estimators': 300, 'max_depth': 8, 'learning_rate': 0.05, 'gamma': 0.1, 'colsample_bytree': 0.8}
LightGBM	{'subsample': 0.8, 'reg_lambda': 1, 'num_leaves': 31, 'n_estimators': 100, 'min_child_samples': 30, 'max_depth': 5, 'learning_rate': 0.05, 'colsample_bytree': 0.7}
CatBoost	{'iterations': 400, 'depth': 6, 'learning_rate': 0.05, 'l2_leaf_reg': 3, 'bagging_temperature': 0, 'random_strength': 1}
TabNet	{'n_steps': 3, 'n_d': 32, 'n_a': 16, 'gamma': 1.0, 'lambda_sparse': 0.001, 'momentum': 0.1, 'optimizer_fn': Adam, 'optimizer_params': {'lr': 0.01}, 'scheduler_fn': StepLR, 'scheduler_params': {'step_size': 10, 'gamma': 0.9}, 'mask_type': 'sparsemax'}

Table 15: Experiment 2: Optimal Hyperparameters for Each Model

Model	**Best Hyperparameters**
Logistic Regression	{'C': 0.01, 'class_weight': 'balanced', 'multi_class': 'multinomial'}
Random Forest	{'criterion': 'gini', 'max_depth': None, 'min_samples_leaf': 1, 'n_estimators': 50}
SVM	{'C': 1, 'class_weight': 'balanced', 'gamma': 'scale', 'kernel': 'linear'}
XGBoost	{'subsample': 0.9, 'reg_lambda': 15, 'n_estimators': 300, 'max_depth': 10, 'learning_rate': 0.07, 'gamma': 0.2, 'colsample_bytree': 0.9}
LightGBM	{'subsample': 0.8, 'reg_lambda': 1, 'num_leaves': 15, 'n_estimators': 100, 'min_child_samples': 30, 'max_depth': 7, 'learning_rate': 0.05, 'colsample_bytree': 0.7}
CatBoost	{'iterations': 400, 'depth': 6, 'learning_rate': 0.05, 'l2_leaf_reg': 3, 'bagging_temperature': 0, 'random_strength': 1}
TabNet	{'n_steps': 5, 'n_d': 32, 'n_a': 8, 'gamma': 1.0, 'lambda_sparse': 0.0001, 'momentum': 0.02, 'optimizer_fn': Adam, 'optimizer_params': {'lr': 0.01}, 'scheduler_fn': None, 'scheduler_params': None, 'mask_type': 'entmax'}

5.3 *Improved Models*

Following hyperparameter tuning, all models were retrained using their optimal configurations and evaluated on the validation set. This evaluation

phase was critical to assess how well the models generalized to unseen data after optimization. For both experiments, performance was assessed using macro-averaged metrics, including F1 Score, Precision, Recall, and AUC. This approach ensured a balanced evaluation across all mental health risk categories, rather than favoring the majority class.

For Experiment 1 (Manual Categorization) (Table 16), tree-based models such as LightGBM and CatBoost continued to show competitive results. However, even after tuning, improvements in F1 Macro remained modest, suggesting that while manual categorization supports interpretability, it may limit the capture of nuanced patterns in the data. Among the models, CatBoost achieved the highest F1 Macro score, followed closely by SVM and Logistic Regression. These results reinforce the strong predictive capabilities of CatBoost when working with structured categorical data.

In Experiment 2 (Sentence Embeddings) (Table 17), despite the use of advanced models, Logistic Regression outperformed more complex classifiers in terms of F1 Macro score. This outcome suggests that the dimensionality-reduced sentence embeddings provided sufficiently rich representations for a linear model to effectively separate classes. Tree-based models like XGBoost and LightGBM performed respectably but did not surpass the simpler Logistic Regression model.

Across both experiments, model evaluation on the validation set revealed that although complex models have the theoretical advantage of modeling non-linear relationships, their effectiveness is highly dependent on the quality and representation of features. Particularly in Experiment 2, the semantic richness captured by sentence embeddings allowed even simple models to perform competitively.

Based on the validation results, the best-performing models were subjected to SHAP-based interpretability analysis and detailed error analysis to further investigate their predictive behavior, before being evaluated on the independent test set.

- Experiment 1: CatBoost on the *Raw* feature set
- Experiment 2: Logistic Regression on the *RFECV_XGB* feature set

Table 16: Experiment 1: Evaluation Metrics (Manual Categorization)

Model	**F1 Score**	**Precision**	**Recall**	**AUC**	**Accuracy**
Logistic Regression (RFECV_XGB)	0.5091	0.5065	0.5147	0.7001	0.5439
Random Forest (RFECV_XGB)	0.4559	0.4778	0.4811	0.6881	0.5543
SVM (RFECV_XGB)	0.5128	0.5132	0.5124	0.7036	0.5454
XGBoost (RFECV_XGB)	0.4740	0.4863	0.4841	0.6820	0.5464
LightGBM (RFECV_XGB)	0.5028	0.5014	0.5059	0.6950	0.5325
CatBoost (Raw)	**0.5172**	0.5148	0.5223	0.7097	0.5483
TabNet (RFECV_XGB)	0.4442	0.4572	0.4508	0.6282	0.5156

Table 17: Experiment 2: Evaluation Metrics (Embeddings)

Model	F1 Score	Precision	Recall	AUC	Accuracy
Logistic Regression (RFECV_XGB)	**0.5114**	0.5089	0.5161	0.6979	0.5459
Random Forest (RFECV_XGB)	0.4344	0.4490	0.4549	0.6673	0.5285
SVM (RFECV_XGB)	0.5029	0.5031	0.5027	0.6999	0.5374
XGBoost (RFECV_XGB)	0.4786	0.4888	0.4879	0.6876	0.5473
LightGBM (RFECV_XGB)	0.5054	0.5031	0.5119	0.7010	0.5394
CatBoost (Raw)	0.4719	0.5205	0.4997	0.7151	0.5736
TabNet (RFECV_XGB)	0.4589	0.4586	0.4627	0.6484	0.5032

5.4 *Model Interpretability and Error Analysis*

5.4.1 *Experiment 1*

In Experiment 1, CatBoost was selected as the best-performing model. Given its capability to handle categorical features effectively, interpretability was explored through built-in feature importance metrics and SHapley Additive exPlanations (SHAP) to understand the model's decision-making process.

As shown in Figure 10 identified *GADE*, employment status , anxiety-related variables and gaming behaviors, such as *Playstyle* as the most influential predictors of mental health risk. These findings suggest that a combination of psychological well-being indicators and lifestyle factors play a central role in mental health classification.

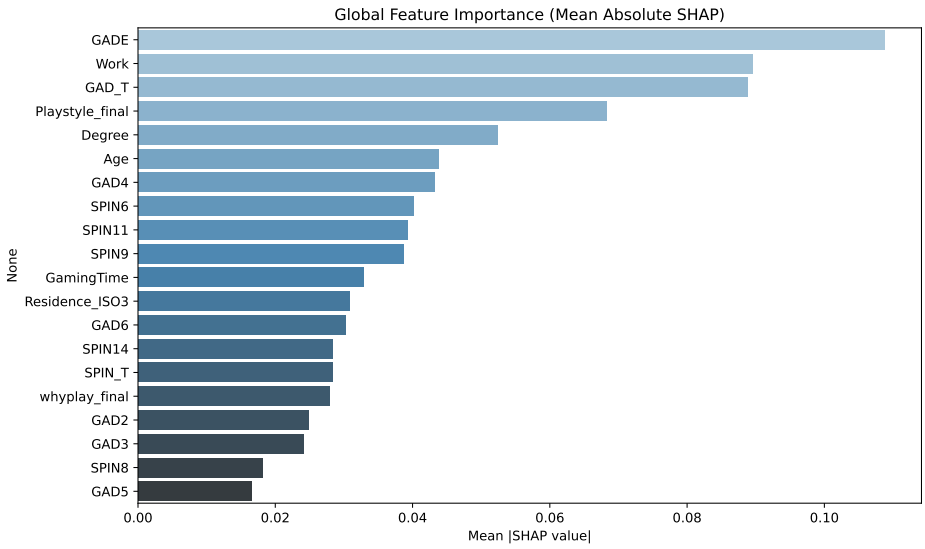

Figure 10: CatBoost - Global SHAP Feature Importance. Mean absolute SHAP values were used to evaluate the overall contribution of each feature to the model's predictions. Higher values indicate greater influence on the model output.

Figure 11 and Table 18 present the confusing matrix and performance metrics. CatBoost demonstrated strong predictive accuracy for the *Low* and *High Risk* categories but struggled to differentiate the *Moderate Risk* group, leading to frequent misclassifications into adjacent categories.

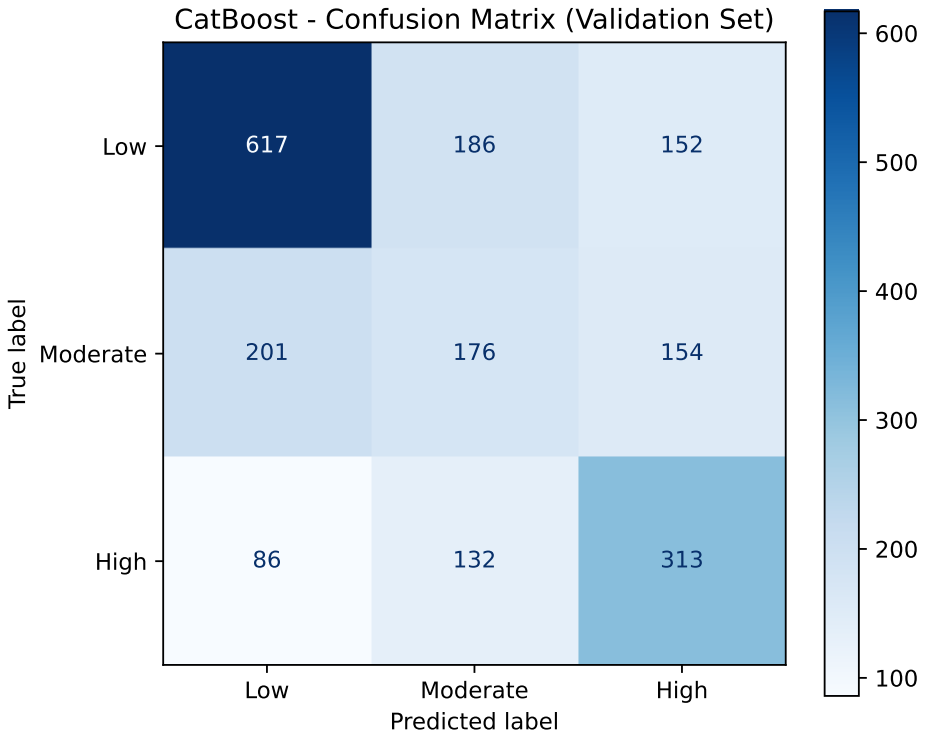

Figure 11: CatBoost - Confusion Matrix (Validation Set). The matrix visualizes the distribution of predictions across the three mental health risk classes. Diagonal cells indicate correct classifications, while off-diagonal values show misclassifications.

Table 18: CatBoost - Classification Report (Validation Set)

Class	Precision	Recall	F1-Score
Low	0.68	0.65	0.66
Moderate	0.36	0.33	0.34
High	0.51	0.59	**0.54**
Accuracy			0.55
Macro Avg	0.51	0.52	0.52
Weighted Avg	0.55	0.55	**0.55**

The ROC curves in Figure 12 further highlight this trend. While the model achieved good separation for the *Low Risk* (AUC = 0.75) and *High*

Risk (AUC = 0.78) groups, the *Moderate Risk* group lagged behind with an AUC of 0.59, reinforcing the observed difficulty in accurately identifying this segment.

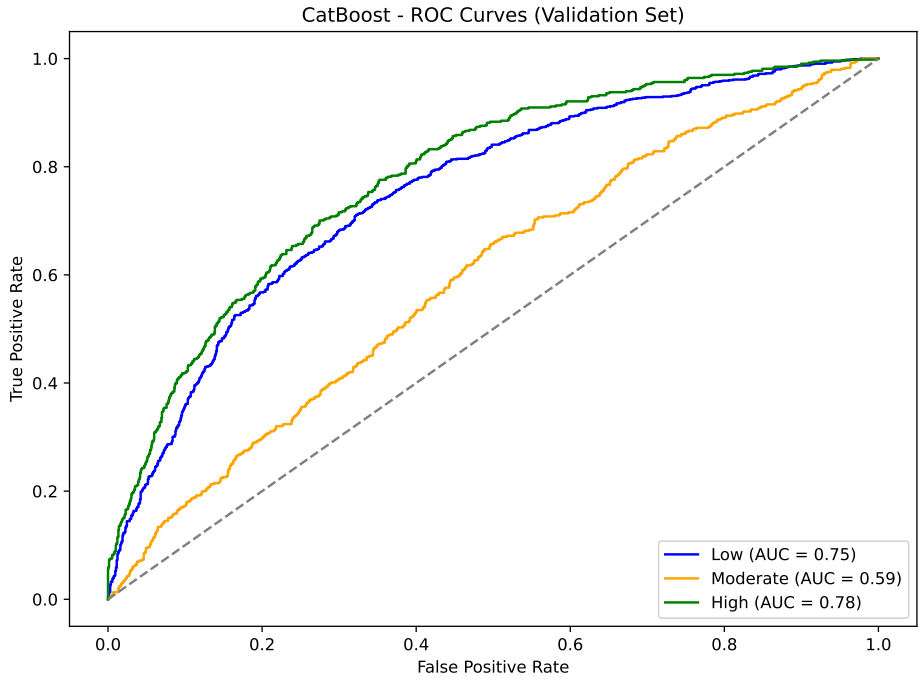

Figure 12: ROC Curve for CatBoost (Validation Set). The ROC curves display the true positive rate versus false positive rate for each mental health risk class using one-vs-rest strategy. AUC values indicate the model's ability to distinguish between classes.

5.4.2 *Experiment 2*

To further investigate the predictive performance and interpretability of the best-performing model in Experiment 2, Logistic Regression, SHAP analysis was performed on the validation set.

Figure 13 presents the global feature importance, highlighting that employment status, function impairment (*GADE*), and psychological factors such as *GAD* and *SPIN* scores had the highest influence on predictions. Notably, sentence embeddings played a secondary role, suggestion that demographic and psychological variables remained the most informative.

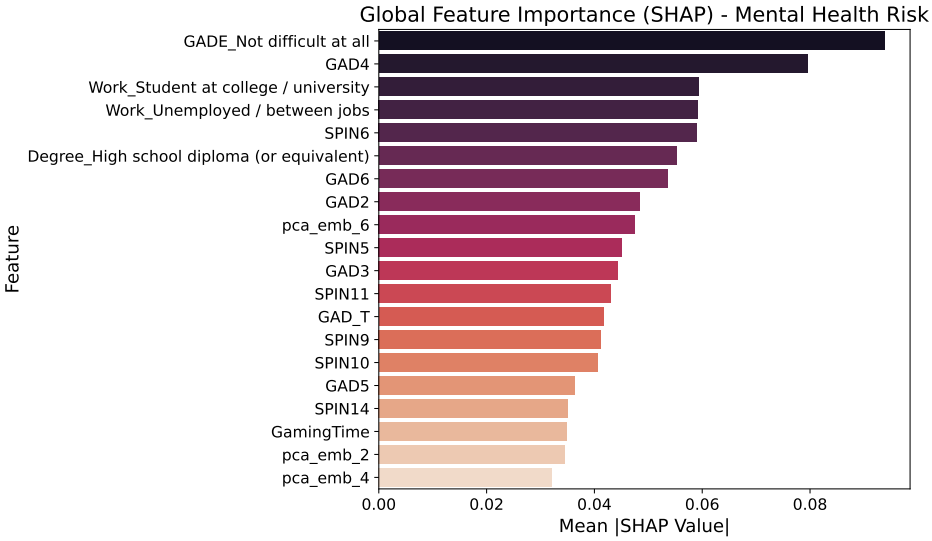

Figure 13: Logistic Regression - Global SHAP Feature Importance. Mean absolute SHAP values were used to evaluate the overall contribution of each feature to the model's predictions. Higher values indicate greater influence on the model output.

Error Analysis

The confusion matrix Figure 14 and Table 19 illustrate the classification performance on the validation set. The model most accurately identified the *Low Risk* group, while the *Moderate Risk* group was the most difficult to classify.

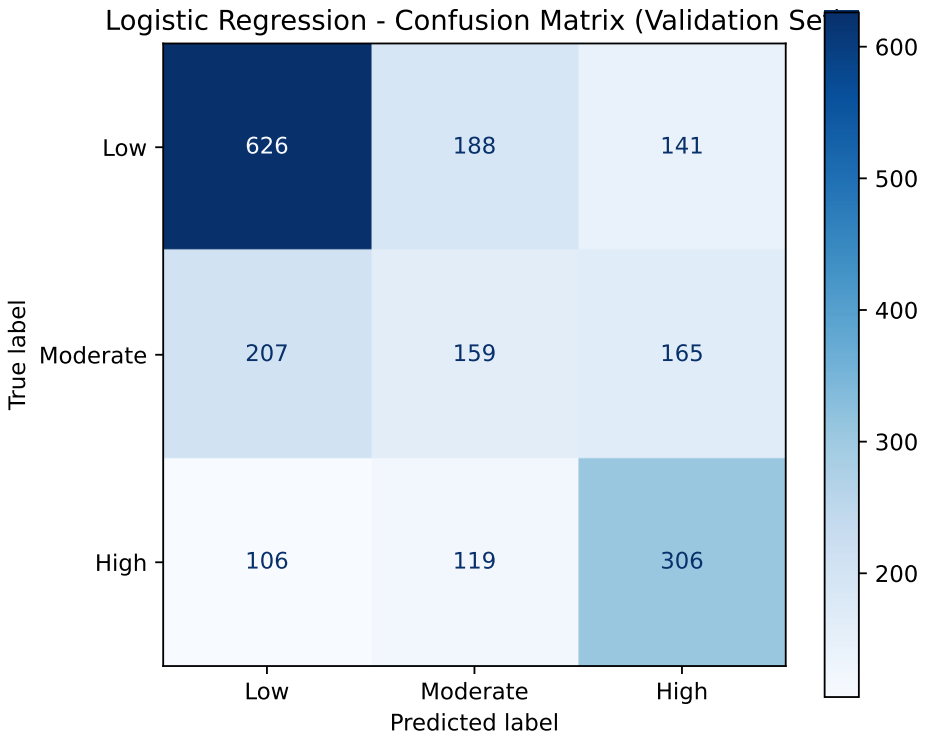

Figure 14: Logistic Regression - Confusion Matrix (Validation Set). The matrix visualizes the distribution of predictions across the three mental health risk classes. Diagonal cells indicate correct classifications, while off-diagonal values show misclassifications.

Table 19: Logistic Regression - Classification Report (Validation Set)

Class	**Precision**	**Recall**	**F1-Score**
Low	0.67	0.66	0.66
Moderate	0.34	0.30	0.32
High	0.50	0.58	0.54
Accuracy			0.54
Macro Avg	0.50	0.51	0.51
Weighted Avg	0.54	0.54	0.54

The ROC curves (Figure 15) summarize the model's ability to distinguish between classes. The model achieved a macro-average ROC AUC of 0.6979, confirming its reliable performance for *Low* and *High Risk* groups, with limited differentiation capability for the *Moderate* group.

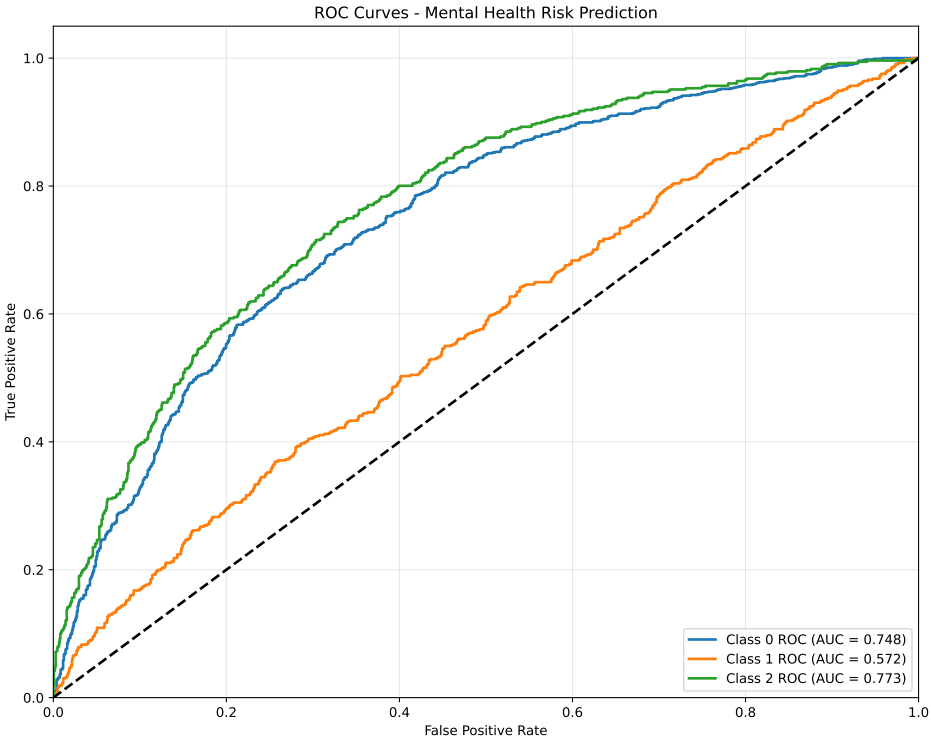

Figure 15: ROC Curve for Logistic Regression (Validation Set). The ROC curves display the true positive rate versus false positive rate for each mental health risk class using one-vs-rest strategy. AUC values indicate the model's ability to distinguish between classes.

5.5 *Final Evaluation on Test Set*

The final evaluation was conducted on the independent test set to assess the generalization performance of the selected models from both experiments.

For Experiment 1 (Manual Categorization), CatBoost achieved an F1 Macro score of 0.4882 and an AUC of 0.6928. As shown in Figure 16, the model maintained good performance in correctly identifying the Low and High Risk categories but continued to struggle with the Moderate Risk group, achieving a recall of 0.29 for this class.

For Experiment 2 (Sentence Embeddings), Logistic Regression achieved a comparable F1 Macro score of 0.4824 and an AUC of 0.6916. As illustrated in Figure 17 and Table 23, this model also demonstrated balanced performance for the Low and High Risk groups but exhibited limited capability in correctly classifying the Moderate Risk category (recall 0.26).

Table 20: CatBoost – Test Set Evaluation Metrics

Metric	Score
Accuracy	0.5201
Precision (Macro)	0.4856
Recall (Macro)	0.4925
F1 Score (Macro)	0.4882
AUC OvR	0.6928

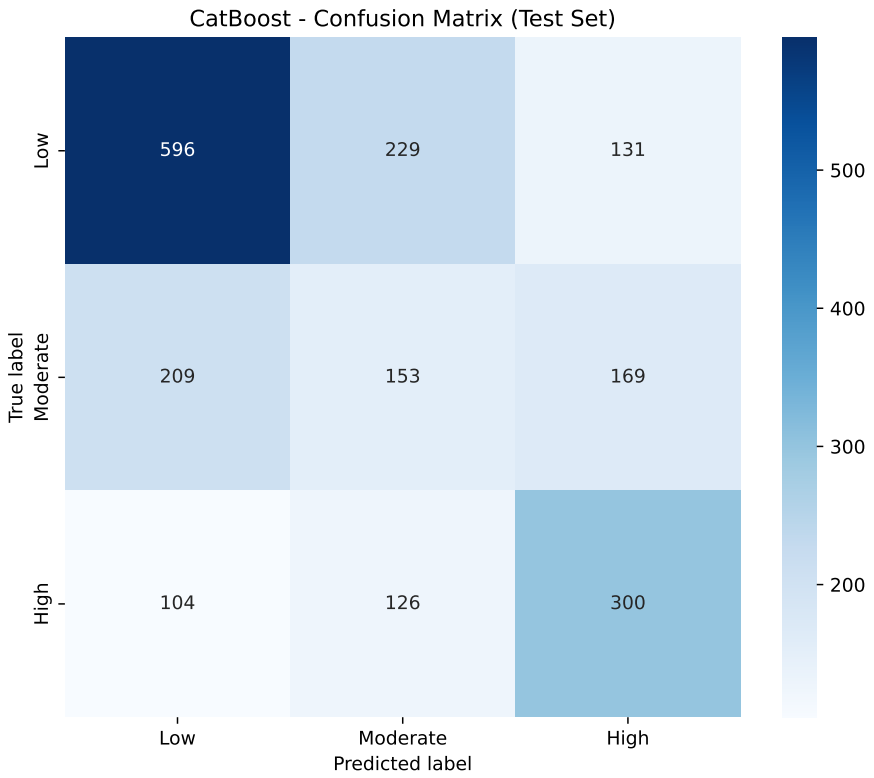

Figure 16: CatBoost – Confusion Matrix (Test Set)

Table 21: CatBoost – Classification Report (Test Set)

Class	Precision	Recall	F1-Score
Low	0.66	0.62	0.64
Moderate	0.30	0.29	0.29
High	0.50	0.57	0.53
Accuracy			0.52
Macro Avg	0.49	0.49	0.49
Weighted Avg	0.52	0.52	0.52

Table 22: Logistic Regression – Test Set Evaluation Metrics

Metric	Score
Accuracy	0.5191
Precision (Macro)	0.4791
Recall (Macro)	0.4888
F1 Score (Macro)	0.4824
AUC OvR	0.6916

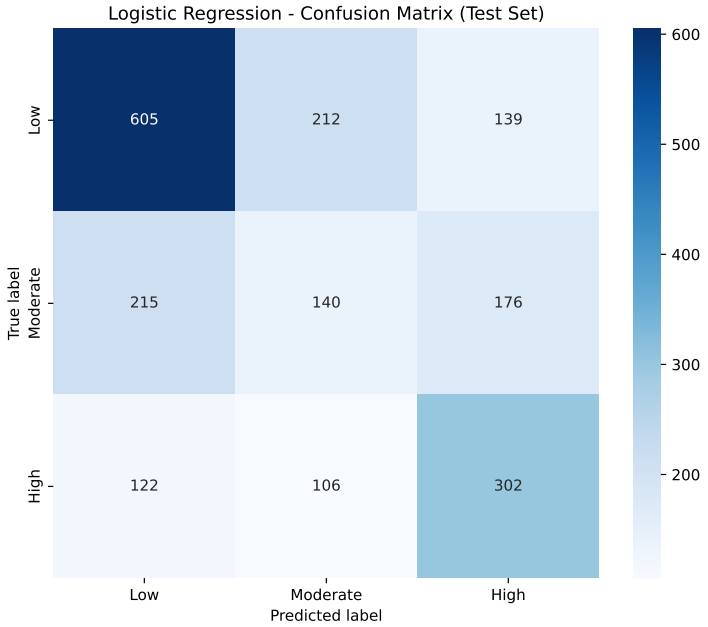

Figure 17: Logistic Regression – Confusion Matrix (Test Set)

Table 23: Logistic Regression – Classification Report (Test Set)

Class	Precision	Recall	F1-Score
Low	0.67	0.66	0.66
Moderate	0.34	0.30	0.32
High	0.50	0.58	0.54
Accuracy			0.54
Macro Avg	0.50	0.51	0.51
Weighted Avg	0.54	0.54	0.54

6 DISCUSSION

The primary goal of this thesis was to evaluate to what extent machine and deep learning models can predict mental health risk levels among video gamers, based on self-reported psychological assessments, gaming behaviors, and demographic factors. Additionally, the study aimed to assess whether the choice of feature representation of open-ended responses - manual categorization or sentence embeddings - affects predictive performance.

6.1 *Results Discussion*

The results demonstrate that, although all models achieved moderate predictive performance, challenges remain in accurately classifying the *Moderate Risk* group. In Experiment 1 (manual categorization), CatBoost achieved the highest F1 Macro score of 0.4882 and an AUC of 0.6928 on the test set. In Experiment 2 (sentence embeddings), Logistic Regression yielded comparable results with an F1 Macro score of 0.4824 and an AUC of 0.6916. Across both experiments, models performed well in identifying low and high-risk groups but consistently struggled to correctly classify the moderate-risk category.

Despite the application of advanced models and hyperparameter tuning, performance remained relatively modest, highlighting the complexity of predicting mental health outcomes from self-reported data.

Interestingly, the simpler Logistic Regression model outperformed more complex models such as XGBoost and TabNet when using sentence embeddings. This suggest that the information captured by the embeddings, after dimensionality reduction with PCA, was sufficiently linearly separable for a simpler model to achieve competitive performance. Additionally, simpler models like Logistic Regression may have generalized better than more complex models, which were more prone to overfitting the semantic variance alongside with other features. These results aligns with previous studies indicating that simpler models often perform robustly in psychological data contexts (Inamdar et al., 2023; Tate et al., 2020).

The under-performance of TabNet across both experiments contradicts the claims of Arik and Pfister (2021) and H. V. Nguyen and Byeon (2023), who reported state-of-art results using TabNet on tabular mental health data. A possible explanation is the relatively moderate sample size and limited variability in certain features, which may have restricted the model's ability to fully leverage its representation learning capabilities.

Regarding the impact of feature representation, models using manually categorized features (Experiment 1) achieved comparable performance

to those using sentence embeddings (Experiment 2). Contrary to the expectations and prior findings (Le Glaz et al., 2021; T. Zhang et al., 2022), sentence embeddings did not improve classification performance. A plausible explanation is that psychological and demographic features held the strongest predictive power, as evidenced by SHAP analyses in both experiments.

SHAP analysis revealed that psychological distress indicators (*GADE*, *GAD_T)*, employment status (*Work*), educational level (*Degree*), and notably, *Playstyle* ranked among the top predictors in Experiment 1, confirming the influence of psychological and gaming-related factors (Bowman et al., 2022; Sauter et al., 2021). In Experiment 2, psychological and socio-economic features as in Experiment 1 remained dominant predictors. The sentence embeddings, derived from gaming motivations responses though, showed comparatively lower but still relevant contribution. One potential reason is that the brevity and diverse nature of open-text answers constrained the complete effectiveness of sentence embeddings, despite their theoretical capacity to extract complex semantic information (Hao et al., 2019).

Moreover, despite using a moderate-balanced dataset and stratified sampling, all models consistently struggled to differentiate the *Moderate Risk* group. This outcome highlights a common challenge in mental health prediction studies, where individuals in moderate states of psychological well-being exhibit overlapping characteristics with both low and high-risk groups (Hasan et al., 2024).

Finally, it is also important to emphasize that although the original target variable *SWL_T* was continuous, this research opted for a classification approach. This decision was based on methodological choices found in the literature (Mahesh et al., 2023; Sau & Bhakta, 2019), where classifying mental health risk levels facilitates interpretability and practical intervention strategies. However, this transformation from a continuous to a categorical target may have introduced information loss, contributing to the modest performance observed.

6.2 *Relevance*

From a scientific perspective, this thesis contributes to the growing body of research on machine learning applications in mental health by applying state-of-the-art models to a large, diverse dataset of over 13,000 gamers. It advances prior work by systematically comparing traditional and advanced models across different feature representations and applying multiclass classification rather than the more commonly used binary approach. Additionally, the use of explainable AI techniques (SHAP) provides critical

insights into the most influential predictors of mental health risk, supporting more transparent and ethically responsible modeling practices.

Societically, the findings highlight the potential for scalable, data-driven solutions for early identification of mental health risk within gaming populations. While the predictive performance achieved in this study does not yet meet the thresholds required for clinical application, it underscores the value of integrating behavioral and psychological data for mental health monitoring in digital environments. These results can inform future developments of mental health screening tools embedded in gaming platforms or online communities, offering opportunities for early intervention before severe psychological issues emerge

6.3 *Limitations and Future Work*

Several limitations should be acknowledged. First, while the dataset was large and diverse, the target variable transformation from a continuous SWL score to categorical risk levels may have reduced the granularity of the predictions. Future research should explore regression-based modeling to predict continuous mental health scores directly. Although classification was preferred for its clinical interpretability, regression models might provide more nuanced insights and support hybrid approaches that combine continuous predictions with actionable classification thresholds.

Second, the open-text responses used for feature engineering were often brief and lacked the emotional richness required for effective sentiment and semantic analysis. Future research should consider employing advanced NLP models fine-tuned on psychological datasets to better capture subtle emotional cues.

Third, while multiple machine learning models were evaluated, only the RFECV_XGB feature set was used for hyperparameter tuning due to time constraints. Exploring a broader range of feature sets during tuning may have led to better model configurations.

Finally, although SHAP analysis provided valuable interpretability insights, only the best-performing models were analyzed in depth. Further research should explore model interpretability across a wider range of models to identify consistent patterns and potential sources of bias.

Future work should also consider exploring longitudinal data to predict mental health trajectories over time, incorporating behavioral data beyond self-reports, and evaluating the potential of ordinal classification approaches to better handle the continuum of mental health states.

7 CONCLUSION

This section provides answers to the research questions formulated in this study.

7.1 *How do traditional models (Logistic Regression, Random Forest, SVM) compare to advanced ensemble models (XGBoost, CatBoost, LightGBM) and deep learning architectures (TabNet) in predicting mental health risk levels?*

Across both experiments, traditional models, achieved comparable performance to advanced ensemble and deep learning models. In Experiment 1 (Manual Categorization), CatBoost slightly outperformed other models with as F1 score of 0.4882. However, in Experiment 2 (Sentence Embeddings), Logistic Regression achieved the highest F1 score of 0.4824, outperforming more complex models. TabNet consistently underperformed in both experiments, suggesting that its deep learning architecture was not well-suited to this task and dataset.

7.2 *What are the most influential features for predicting mental health risk categories, as identified through SHAP analysis and feature selection techniques?*

Both SHAP analysis and feature importance methods identified psychological indicators (*GADE*, *GAD_T*), along with socio-economic factors like employment status and education level as the strongest predictors of mental health risk.

7.3 *Which model provides the best balance between Precision and Recall specifically for identifying individuals at high mental health risk, based on error analysis?*

Both CatBoost and Logistic Regression provided a balanced performance for identifying individuals at high risk. On the test set, Logistic Regression achieved a precision of 0.50 and recall of 0.58 for the high-risk group, while CatBoost achieved similar values (precision 0.50, recall 0.57). However, both models struggled to accurately classify the moderate-risk group, highlighting a persistent challenge in detecting early signs of psychological distress.

7.4 *Overarching RQ: To what extent can machine learning models predict mental health risk levels among video gamers based on self-reported gaming behaviors and psychological assessments, and how does the choice of feature representation (manual categorization vs. sentence embeddings) affect predictive performance?*

Machine learning models achieved moderate effectiveness in predicting mental health risk levels among video gamers. Although both manual feature categorization and sentence embeddings were explored, neither approach demonstrated clear superiority in predictive accuracy. Psychological and socio-economic variables remained dominant predictors, but gaming motivations also played a relevant role. Despite applying advanced models and hyperparameter optimization, accurately identifying individuals at moderate risk remains a persistent challenge.

Overall, this study emphasizes the promise of machine learning in assessing mental health risks, while also highlighting the necessity for further improvements, which future research should address based on the provided recommendations.

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APPENDIX A

APPENDIX A: SOFTWARE AND PACKAGES

The data cleaning, preprocessing, and modeling were performed in **Python Version 3.11.12**. The packages and libraries used were:

- **Pandas** (Version 2.2.2) (pandas development team, 2020)
- **NumPy** (Version 2.0.2) (Harris et al., 2020)
- **Scikit-learn** (Version 1.6.1) (Pedregosa et al., 2011)
- **SciPy** (Version 1.15.3) (Virtanen et al., 2020)
- **Matplotlib** (Version 3.10.0) (Hunter, 2007)

- **Seaborn** (Version 0.13.2) (Waskom, 2021)
- **XGBoost** (Version 2.1.4) (T. Chen & Guestrin, 2016)
- **LightGBM** (Version 4.5.0) (Ke et al., 2017)
- **CatBoost** (Version 1.2.8) (Prokhorenkova et al., 2018)
- **SHAP** (Version 0.47.2) (Lundberg & Lee, 2017)
- **Sentence-Transformers** (Version 4.1.0)
- **Pytorch TabNet**

APPENDIX B

APPENDIX B: MISSING DATA

Table 24: Missing Data Summary

Feature	Missing Count	Missing Percentage
League	1852	13.76%
Degree	1577	11.71%
SPIN_T	650	4.83%
GADE	649	4.82%
SPIN13	187	1.39%
SPIN11	187	1.39%
SPIN17	175	1.30%
SPIN12	168	1.25%
SPIN5	166	1.23%
SPIN10	160	1.19%
SPIN4	159	1.18%
SPIN9	158	1.17%
SPIN14	156	1.16%
SPIN6	156	1.16%
SPIN2	154	1.14%
SPIN16	147	1.09%
SPIN15	147	1.09%
SPIN8	144	1.07%
SPIN3	140	1.04%
SPIN7	138	1.02%
SPIN1	124	0.92%
Birthplace_ISO3	121	0.90%
Residence_ISO3	110	0.82%
streams	100	0.74%
Work	38	0.28%
Hours	30	0.22%

APPENDIX C

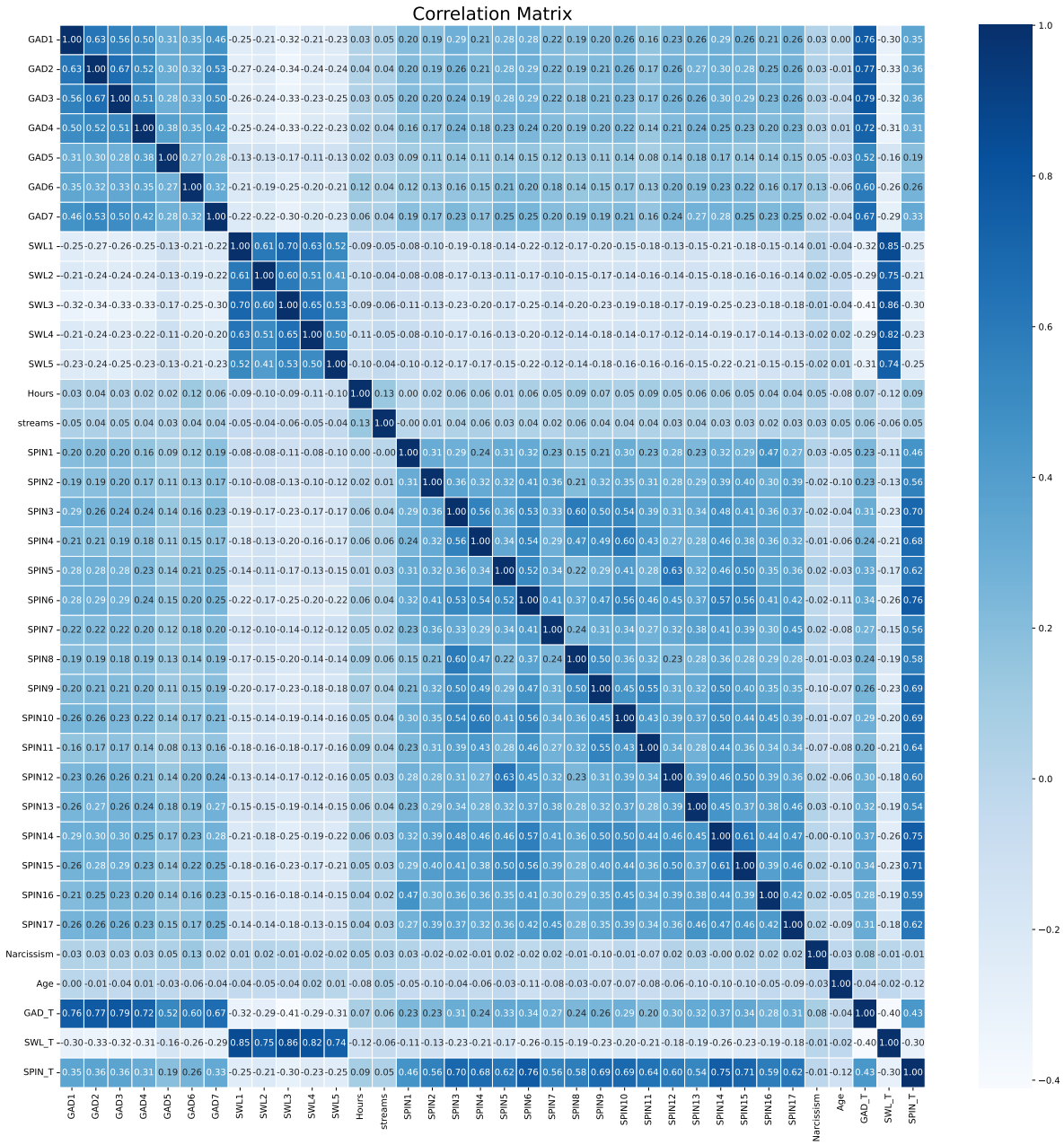

Figure 18: Full Correlation Matrix of all numerical Features. Visualizes the strength and direction of pairwise associations, with darker colors indicating stronger correlation (positive or negative).

APPENDIX D

Table 25: Top 10 Features Ranked by ANOVA F-Test Scores for both Experiments. Higher F-scores indicate stronger discriminatory power in predicting mental health risk levels.

Feature	**ANOVA F-score**	**ANOVA p-value**
GAD_T	795.52	0.000
GAD2	552.54	0.000
GAD3	525.62	0.000
GAD4	502.25	0.000
GAD1	475.81	0.000
GADE_Not difficult at all	431.48	0.000
SPIN_T	427.60	0.000
GAD7	400.30	0.000
GAD6	347.11	0.000
SPIN14	332.59	0.000